

Role Discovery Under Data Scarcity: Unsupervised Player Profiling in the Premier League After the Withdrawal of Public Advanced Football Statistics

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September 6, 2026

Abstract

In January 2026 the largest free source of advanced football statistics removed every metric derived from its commercial provider, retroactively and without replacement. Expected goals, passing volume, carrying, chance creation and the entire advanced goalkeeping table ceased to be publicly available. We measure what player role structure survives that withdrawal. Merging the statistics that remain with an independent expected goals source yields 18 features for 366 Premier League outfield players, on which we run dimensionality reduction, clustering under a selection rule fixed in advance, bootstrap stability, supervised validation, and a replication on a second season. Three results hold. Correlations reverse sign across position groups for 88 of 153 feature pairs, so pooled correlation analysis describes no real population. The data supports one binary division, attacking against defensive involvement, which cuts across listed positions at an adjusted Rand index of only 0.226 while a supervised model reaches 0.803 accuracy on the same features. That global division survives shrinkage toward the position mean at 0.945, whereas the divisions within defender and forward groups fall to 0.453 and 0.626 and are substantially descriptions of sampling noise. Public football data now supports coarser role analysis than it did a year ago.

1 Introduction

Public football data matured over the last decade to the point where every outfield player in a major league carried a rich statistical profile. Expected goals models [Rathke, 2017, Eggels et al., 2016, Vatvani, 2022] moved from research novelty to broadcast furniture, and free sites published per-90 measures of passing, carrying, pressing and chance creation that a decade earlier existed only inside clubs. Most public analysis nonetheless stopped at leaderboards, ranking players on one statistic at a time rather than asking what combinations of statistics describe a role.

Research on roles has generally required data that is not public. Bialkowski et al. [Bialkowski et al., 2014] used tracking data to recover formations and positional roles from player coordinates. Decroos and Davis [Decroos and Davis, 2019] learned player vectors from event streams, Pappalardo et al. [Pappalardo et al., 2019] built a performance rating from event data across several leagues, and Aalbers and Van Haaren [Aalbers and Van Haaren, 2018] distinguished playing styles from match events. All of these depend on proprietary event or tracking feeds. The question of how much role structure is recoverable from freely available season aggregates has received much less attention, and the answer has just changed sharply.

On 20 January 2026 the operator of FBref [Sports Reference LLC, 2026], the most widely used free source of advanced football statistics, removed every statistic derived from its commercial

provider after the licence was terminated and deletion was demanded. The removal applied retroactively to completed seasons. Expected goals, expected assisted goals, shot and goal creating actions, all touch, carry and take-on counts, all passing volume and progression, and the entire advanced goalkeeping table ceased to be available. What remains is close to the statistical vocabulary of two decades ago.

This paper asks what player role structure survives that withdrawal. We assemble the statistics that remain, merge an independent expected goals source to restore the shot quality block, and run a full role discovery pipeline: dimensionality reduction, clustering with a selection rule fixed in advance, bootstrap stability, supervised validation, goalkeeper and team level extensions, and a cross-season replication. Every figure and every number in this paper is generated by code in an accompanying repository built on the `soccerdata` library [Robberechts, 2026], and no value in the text was transcribed by hand.

Five findings are worth stating in advance.

First, the withdrawal is a methodological trap and not merely a loss of columns. The affected tables are still served with complete headers and one row per player, and only the values are gone. A feature set assembled from the published schema resolves correctly and produces an empty matrix, so feature selection must now be made against observed values rather than documentation.

Second, correlations between statistics reverse sign across position groups. Of 153 feature pairs, 88 change sign between defenders, midfielders and forwards, and all ten of the most divergent pairs do. A pooled correlation matrix therefore averages over opposing relationships and describes no actual population of players.

Third, the surviving data supports a coarse binary division and nothing finer. The selection rule returns two clusters in every scope, and the division it finds is attacking against defensive involvement rather than a rediscovery of listed positions, with an adjusted Rand index of only 0.226 against the position groups. A supervised model reaches an accuracy of 0.803 on the same features, so the weak agreement reflects the clustering choosing a different axis rather than an uninformative feature set.

Fourth, a correction that public football analysis applies routinely is miscalibrated. Defensive counts must be adjusted for how much time a team spends without the ball, and the conventional adjustment assumes the two are directly proportional. Fitted against team data that assumption fails: the elasticities are 0.525, 0.432 and 0.283, so the conventional correction overcorrects by a factor of two to three and, applied here, replaces the confound it removes with a larger one of the opposite sign.

Fifth, the global division is robust while the finer divisions inside position groups are not. Bootstrap stability, cross-season replication and empirical Bayes shrinkage together place the global partition at an adjusted Rand index of 0.945 under shrinkage toward the position mean, while the defender partition falls to 0.453 and fails all three tests. The archetypes that public data can still support are therefore fewer, and coarser, than the ones it supported a year ago.

2 Data

2.1 Sources and retrieval

All statistics are season aggregates for the English Premier League, retrieved for the 2024-25 season, which the study is about, and the 2025-26 season, which is used only for replication. Retrieval goes through the `soccerdata` library, which caches every page to disk and rate limits requests to one every seven seconds, so no request is ever issued twice and the source site is not burdened. Raw pages are cached before any transformation, and every retrieved table is written to disk untouched

Table 1: Populated columns in each FBref player table. The tables are still served with complete headers, so the empty counts are the number of columns that resolve correctly and contain no values.

Table	2024-25		2025-26	
	Columns	Empty	Columns	Empty
standard	17	0	17	0
shooting	11	0	11	0
passing	21	19	21	19
passing types	16	14	16	14
goal shot creation	17	16	17	16
defense	17	14	17	14
possession	21	20	21	20
playing time	17	0	17	0
misc	13	2	13	2
keeper	19	0	19	0
keeper adv	26	23	26	23

before a single column is renamed.

Two properties of the retrieval path are worth recording, because both took work to establish and both affect what any replication of this study will encounter. The first is that FBref sits behind a challenge that returns a refusal to ordinary HTTP clients, so only the library release that drives a real browser can reach it. The second is that this release, while it can reach the site, exposes only five of the eleven season stat tables that the site publishes. We restore the missing six by subclassing the reader and re-declaring the table identifiers, reusing the library’s own downloader, cache and parser rather than writing any request loop of our own. Establishing the correct identifiers required inspecting live pages, which is how we found that the goal and shot creation table is served under an identifier that does not match its documented name.

2.2 The withdrawal of advanced statistics

On 20 January 2026 the operator of FBref removed every statistic derived from its commercial data provider, following termination of the licence and a demand that the data be deleted. The removal was retroactive rather than prospective, so it applies to completed seasons as well as future ones.

The practical consequence is more subtle than a set of missing tables, and it is a trap for anyone building on this source. The affected tables are still served, still carry their full column headers, and still return one row per player. Only the values are gone. A feature list assembled from the site’s published schema therefore looks entirely correct and yields a feature matrix that is empty. Table 1 reports the audit we ran to establish exactly what survived, counting non-null values for every column of every table in both seasons.

What remains is close to the statistical vocabulary of two decades ago: appearances, minutes, goals, assists, shots, shots on target, cards, fouls committed and drawn, offsides, crosses, interceptions and tackles won, together with basic goalkeeping volume. What disappeared is the entire modern apparatus, including expected goals, expected assisted goals, shot and goal creating actions, every touch and carry and take-on count, all passing volume and progression, and, for goalkeepers, post-shot expected goals, sweeper actions, distribution length and cross handling.

2.3 Restoring shot quality

Because a role discovery study with no measure of chance quality would be badly impoverished, we merge a second and independent source. Understat publishes its own expected goals model and continues to provide non-penalty expected goals, expected assists, key passes, shot counts, and the two possession involvement measures xGChain and xGBuildup. These restore the creation and shot quality block that the withdrawal removed, at the cost of introducing a second provider whose model differs from the one that produced the original FBref values. We treat that as a limitation rather than a repair, and we never mix an Understat quantity with an FBref quantity that measures the same thing.

Records are matched across the two sources within club, using three passes of decreasing strictness: exact match on a normalised name, then a token subset rule that resolves short forms against full names, then an approximate string match. The token subset pass is accepted only when exactly one candidate matches, so a club fielding several players who share a given name can never be silently conflated. Seven clubs are named differently by the two sites and are mapped explicitly. The procedure matches 97.0 percent of 2024-25 records and 96.2 percent of 2025-26 records. Unmatched players retain missing values for the Understat block and are counted rather than quietly dropped.

2.4 Filtering, normalisation and feature construction

The 2024-25 pull contains 574 player records, of which 518 are outfield players and 44 are goalkeepers. A player who changed club during the season appears once per club in the source. We retain those rows for reference but conduct every analysis on an aggregated row per player, in which counting statistics are summed and rates are recomputed from their components rather than averaged across spells of unequal length, since averaging a rate over unequal exposure is not the rate. 12 players required this treatment in 2024-25.

Players with fewer than 450 minutes, five full matches, are excluded from clustering while being retained in the descriptive tables. The threshold trades sample size against the noise in a per-90 rate computed over a short exposure, and Appendix B reports the same analysis at 270 and 900 minutes. At the chosen threshold 366 outfield players are eligible in 2024-25 and 372 in 2025-26, distributed across 136 defenders, 181 midfielders and 49 forwards.

Counting statistics are divided by nineties played. Quantities that are already rates, such as shot accuracy, are left alone rather than being divided a second time. The three statistics that can only be recorded while the opponent has the ball, interceptions, tackles won and fouls committed, are additionally corrected for how much time a player’s team spends out of possession, by the procedure set out in Section 3.1. Minutes and nineties are retained as context columns and never used as features, since squad usage describes a player’s opportunity rather than the role he performs within it. For the same reason the team success columns, points per match and goal difference while on the pitch, are excluded from every feature set: they measure the quality of the squad a player happens to belong to.

Positions are parsed from the source’s comma separated string. The first listed position gives a primary position and a position group in defender, midfielder, forward and goalkeeper, while the full string is retained for the analysis of players listed at more than one position. The final outfield feature set contains 18 features and is listed in full in Appendix A. Every feature in it was confirmed to hold non-null values in both seasons before inclusion, which is the only reliable defence against the empty column problem described in Section 2.2. Features are standardised twice, once across the whole outfield pool for the global analysis and once within position group for the position stratified analysis, and both versions are retained.

2.5 Known limitations of the data

Season aggregates discard context. A tackle count does not record where on the pitch the tackles happened, an expected goals total does not distinguish a striker who takes six low quality shots from one who takes two good ones, and the possession correction described in Section 3.1 reaches only the three statistics recorded while the opponent has the ball, leaving the attacking statistics still sensitive to how much of it a player’s team enjoys. Per-90 rates computed over a partial season remain noisy even above the minutes threshold, which is the motivation for the shrinkage analysis in Section 4.13.3. The study covers one league and two seasons, so nothing here should be read as a claim about football in general.

3 Methods

Every procedure below is seeded from a single constant, and every hyperparameter is declared in one configuration file that the analysis code reads. Nothing is tuned after seeing a result. All computation uses scikit-learn [Pedregosa et al., 2011] unless stated otherwise.

3.1 Possession adjustment

Defensive actions can only occur while the opponent has the ball. A player in a side that concedes possession therefore accumulates interceptions, tackles and fouls partly through exposure rather than through role or ability, and since possession correlates with team strength this confound would enter the feature set as a proxy for the quality of a player’s team. Team possession share survived the withdrawal even though the touch counts it is derived from did not, which is what makes a correction possible at all.

The conventional correction multiplies a defensive rate by the ratio of a reference opponent possession to the actual one, which is equivalent to assuming that defensive volume is directly proportional to time spent out of possession. That is an empirical claim rather than a definition, and it can be tested. Regressing the logarithm of team defensive volume per ninety on the logarithm of opponent possession across the 40 team-seasons available gives the elasticity, which is the percentage change in defensive volume produced by a one percent change in opponent possession. Direct proportionality corresponds to an elasticity of one.

The fitted elasticities are 0.525 for interceptions, 0.432 for tackles won and 0.283 for fouls committed. All three are well below one, so defensive volume rises with opponent possession, confirming that the confound is real, but far less than proportionately. A side that spends more time without the ball does not simply make proportionally more tackles: it also defends deeper, concedes territory without contesting it, and its defensive actions saturate. Applying the conventional exponent of one would therefore overcorrect by a factor of roughly two to three, and we verified that it does: under that exponent the correlation between defensive rates and team possession does not fall toward zero but changes sign and grows, so the adjustment injects more team dependence than it removes.

We therefore adjust with the measured elasticity rather than the assumed one. Each defensive count is multiplied by the exposure ratio raised to the power of its own fitted elasticity, which leaves a player at a club with league-average possession unchanged and corrects the others by the amount the data says exposure actually matters. The exponent is estimated from team totals rather than player rates, because the quantity being corrected is a team-level exposure effect and player rates carry position and role variation that would swamp it. It is pooled across seasons because twenty clubs give a noisy slope, and because the exponent is a nuisance parameter that uses no player-level or cluster-level information, so a shared value cannot manufacture the cross-season agreement that

Section 4.13 measures. Attacking statistics are deliberately left unadjusted, since they scale with a team’s own possession rather than the opponent’s and correcting both directions with the same instrument would double count.

3.2 Dimensionality reduction

Principal component analysis is fitted to the standardised feature set, both globally and separately within each position group. The global embedding is dominated by the axis that separates defenders from forwards, which is the least interesting distinction available because it is already recorded in the listed position, so the role profiling that follows uses the position stratified fits.

Uniform manifold approximation and projection [McInnes et al., 2018] is fitted over a grid of twelve configurations, with the number of neighbours in $\{10, 15, 30, 50\}$ crossed with a minimum distance in $\{0.0, 0.1, 0.5\}$. Configurations are ranked by trustworthiness, which measures the degree to which points that are close in the embedding were also close in the original space, and the full grid is reported in Appendix C. Selection by an explicit criterion matters here because a manifold method will produce a visually appealing picture at almost any setting.

t-distributed stochastic neighbour embedding [van der Maaten and Hinton, 2008] is computed at a perplexity of 30 for visual contrast only. It is never used as an input to clustering, because its objective preserves local neighbourhoods at the expense of the global geometry that a distance based clustering algorithm depends upon.

The three embeddings are compared on trustworthiness, on continuity, which is trustworthiness computed with the roles of the two spaces exchanged, and on a nearest neighbour position purity score at $k = 10$ that reports the share of a player’s ten nearest neighbours that share his listed position group.

3.3 Clustering

Clusters are estimated with k -means as the primary method, with a Gaussian mixture model whose Bayesian information criterion is reported across the same range of component counts, and with HDBSCAN [Campello et al., 2013], for which we report the fraction of players assigned to noise. Each is fitted both to the standardised features and to the principal component space retaining ninety percent of variance, and each reported result states which space produced it. Clustering is performed globally and within each of the three outfield position groups.

The number of clusters is selected by a rule fixed in advance of any fitting, which matters because four criteria applied after the fact can be made to justify almost any choice. Over k from 2 to 12 we compute the silhouette coefficient [Rousseeuw, 1987], the Davies-Bouldin index, the Calinski-Harabasz index, and the gap statistic [Tibshirani et al., 2001]. The gap statistic has no implementation in the library we use, so it is implemented here against the original description, comparing the within-cluster dispersion to its expectation under fifty reference datasets drawn uniformly over the bounding box of the data in the fitted space. The rule selects the smallest k whose silhouette lies within 0.02 of the best silhouette observed and which ranks in the top three on at least two of the silhouette, Calinski-Harabasz and gap criteria, with Davies-Bouldin breaking ties. Which criteria agreed, and which did not, is reported alongside the choice.

3.4 Stability and validation against listed positions

A partition can be produced from any dataset, so the question that matters is whether it survives resampling. We draw two hundred bootstrap resamples, refit the clustering on each, and compute the adjusted Rand index [Hubert and Arabie, 1985] between each bootstrap partition and the reference

partition on the players common to both. We also accumulate a consensus matrix recording, for every pair of players, the fraction of resamples in which the two were assigned together. Clusters whose members co-assign below 0.6 on average are reported as unstable, and they are described as such in the results rather than being renamed or merged to make the account tidier.

The relationship between the discovered clusters and the listed positions is quantified by normalised mutual information and by the adjusted Rand index, computed against both the four-way position group and the full position string. Players whose cluster assignment disagrees with their listed group, and players whose Gaussian mixture posterior is split across components, are extracted as candidate hybrids.

3.5 Archetype profiling

For every retained cluster we compute the centroid profile as a standardised deviation from the mean of its position group, take the eight features with the largest absolute deviation as the distinguishing features, and list the ten players closest to the centroid and three players near the boundary. Archetype names are assigned only after inspecting these profiles and the players they contain, and each name is accompanied by a justification that cites the distinguishing features and the example players. The names are hypotheses about what the statistical profile represents, not observations, and they carry no authority beyond the profile that produced them.

Outliers are identified within position group with an isolation forest [Liu et al., 2008], reported alongside a Mahalanobis distance so that a distance-based and a partition-based notion of unusualness can be compared.

3.6 Supervised validation

Unsupervised structure is easy to over-interpret, so we ask a supervised model what separates the listed positions. Logistic regression and gradient boosting are trained to predict the position group from the feature set under five-fold stratified cross validation, and we report accuracy, macro F1 and the confusion matrix. The purpose is interpretability rather than accuracy: SHAP values [Lundberg and Lee, 2017] on the boosted model identify which statistics actually distinguish each position in this feature set. Players the classifier places in the wrong group are cross referenced against the candidate hybrids from the clustering, since agreement between two methods that fail in the same place is more informative than either alone.

A secondary ridge regression predicts non-penalty expected goals per ninety from the non-shooting features only, so that the fitted value represents the shot volume a player’s role would ordinarily generate and the residual represents the part not explained by role. We report the cross validated coefficient of determination and treat the residual ranking with heavy caution, because a residual from a model with modest explanatory power is mostly noise.

3.7 Goalkeepers

The goalkeeper analysis is severely constrained by the withdrawal described in Section 2.2, which removed post-shot expected goals, sweeper actions, distribution and cross handling. Only shot stopping and workload survive, so we build two composites rather than the five that the surviving column headers appear to promise. In place of post-shot expected goals we regress saves on shots on target faced and use the residual as a shot stopping index. That index adjusts for the volume of work a goalkeeper receives but not for its difficulty, which makes it a substantially weaker instrument, and it is reported as such.

3.8 Team signatures

Each club is summarised in two ways: as a minutes-weighted centroid of its outfield players in the shared principal component space, and as the share of its outfield minutes played by each archetype. Clubs are then clustered hierarchically on archetype composition and compared against final league position.

3.9 Additional analyses

Three further analyses address specific threats to the interpretation.

Player similarity is computed as cosine similarity within position group, and the nearest neighbours of a set of well known players are reported so that a reader with football knowledge can judge whether the geometry corresponds to anything real.

Cross-season replication refits the entire pipeline on the replication season and computes the adjusted Rand index between the two seasons for players present in both, which distinguishes archetypes that describe stable playing roles from artefacts of a single season.

Per-90 rates computed over partial seasons are noisy, and that noise is largest exactly where the minutes are fewest. We therefore apply empirical Bayes shrinkage toward the position group mean, with the degree of shrinkage determined by minutes played [Efron and Morris, 1977], refit the clustering on the shrunken features, and report how many assignments change. A partition that dissolves under shrinkage was substantially a description of sampling noise.

3.10 Reproducibility

The analysis runs from a single command against pinned dependencies and a fixed seed. Every figure and every number in this paper is generated by code in the accompanying repository and read into the document from generated definitions at build time, so no value in the text was transcribed by hand.

4 Results

4.1 What survived the withdrawal

The audit reported in Table 1 establishes the scale of the loss. In the goalkeeping detail table, 23 of 26 statistical columns are entirely empty in both seasons, leaving three: nineties played, goals conceded and penalties conceded. Neither of the two surviving substantive columns supports any of the analyses that table exists to enable. The pattern is the same across the outfield tables, where passing, possession, and goal and shot creation retain their headers and lose their contents.

This is the first result of the paper, and it is a methodological one. A study that selected its features from the published schema of this source would produce an empty feature matrix while every column name resolved correctly, and the failure would surface only as unexplained missingness far downstream. Feature selection against observed values rather than published schemas is now a necessary step when working with public football data.

4.2 Correcting defensive counts for possession

Before any structure is estimated, the defensive counts have to be separated from the opportunity to record them. Across the 40 team-seasons available, team possession ranges from 40.6 to 61.3 percent, and team defensive volume does rise as opponent possession rises, so the confound is real.

Table 2: Possession adjustment. The elasticity is fitted from team totals. The final three columns give the correlation between each per-90 rate and team possession: unadjusted, adjusted with the fitted elasticity, and adjusted with the conventional exponent of one. A value near zero is the goal.

Statistic	Elasticity	Correlation with team possession		
		Unadjusted	Adjusted	Exponent one
Interceptions	0.525	-0.122	-0.022	0.070
Tackles won	0.432	-0.111	-0.018	0.105
Fouls committed	0.283	-0.102	-0.037	0.129

It is much weaker than proportional, however. The fitted elasticities are 0.525 for interceptions, 0.432 for tackles won and 0.283 for fouls committed, against the value of one that the conventional adjustment assumes.

That difference matters, and Table 2 shows why. The three raw correlations with team possession are -0.122, -0.111 and -0.102, all negative, since a player whose team keeps the ball records fewer defensive actions. Applying the conventional exponent overshoots every one of them: the correlations become 0.070, 0.105 and 0.129, reversing sign in all three cases and, for fouls committed, ending further from zero than where it started. The correction passes through the confound and out the other side. Adjusting with the measured elasticity instead leaves -0.022, -0.018 and -0.037, which are three to four times closer to zero than the conventional exponent achieves and are the smallest residual dependence available from either choice.

Two qualifications belong with this. The correction is calibrated on a league whose possession shares are relatively compressed, so the elasticities should not be transported to a competition with a wider spread without refitting. More importantly, adjusting the defensive counts does not make the feature set as a whole team-neutral, because the attacking statistics still scale with a team’s own share of the ball. What follows should be read as having removed one identifiable confound, not all of them.

4.3 Why the correction is one sided

Correcting only the defensive statistics invites an obvious objection. Attacking output also depends on possession, so the feature set is still team dependent and the correction looks like half a correction. The objection is worth testing rather than answering by assertion, and the test produces the sharpest methodological result in the paper.

The two sides of the ball do not behave alike. Fitted the same way, against a team’s own possession rather than the opponent’s, the attacking elasticities are 1.09 for shots, 1.48 for non-penalty expected goals, 1.63 for non-penalty goals, 1.72 for assists, 2.12 for xGChain and 2.44 for xGBuildup. Every one of them except crossing, at 0.42, exceeds one, while every defensive elasticity falls well below it. Figure 1 shows the two families either side of the proportional line.

That asymmetry is not a curiosity, it is the whole argument. For defending, opponent possession is an exposure denominator in the plain sense that a tackle cannot occur while your own side holds the ball, and the sub-proportional response says exposure is exactly what it measures. For attacking, an elasticity above one says the opposite: ten percent more of the ball buys appreciably more than ten percent more output. Returns to possession are increasing. That is a statement about football rather than about measurement, so dividing it out would remove signal along with any bias. The two Understat involvement measures make the point most starkly, since both exceed two: a side that keeps the ball does not merely have more possessions, its players are involved in more of each.

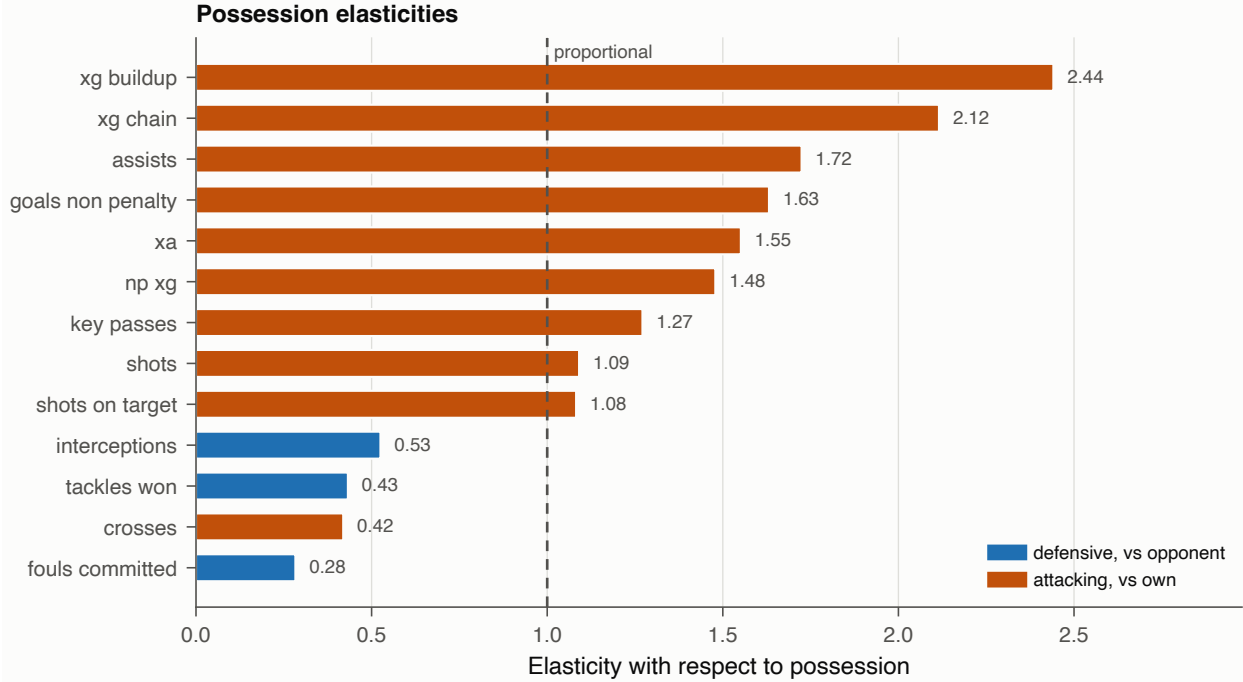


Figure 1: Possession elasticities for both families of statistic, fitted from team totals. Defensive volume responds to opponent possession less than proportionately, while attacking output responds to a team’s own possession more than proportionately. Crossing is the one attacking statistic that behaves like a defensive one.

The counterfactual was nonetheless computed. Applying the fitted exponents to the attacking block as well moves the correlation between club centroids and final league position from -0.861 to -0.610, so a symmetric correction does remove a real part of the team relationship rather than nothing at all. It does not remove all of it, the partition moves at an adjusted Rand index of 0.872 against the one reported here, and the silhouette drifts from 0.285 to 0.281. The choice is therefore a judgement rather than a fact, and it is recorded as one: we treat possession driven productivity as something a role profile should reflect, not something to divide away, and a reader who takes the opposite view has the numbers to see what that would cost.

4.4 Descriptive structure

After filtering, 366 outfield players are eligible in the primary season, comprising 136 defenders, 181 midfielders and 49 forwards. Summary statistics by position group appear in Table 3, and the distributions of four representative statistics are shown in Figure 2.

The feature set is strongly non-normal. All 18 of the 18 features reject the Shapiro-Wilk null at the five percent level, mean absolute skewness is 1.13, and the most skewed feature is goals per shot at 2.89. Mardia’s test rejects multivariate normality decisively on both the skewness and kurtosis components. Principal component analysis requires no distributional assumption to be computed, so this does not invalidate the decomposition, but it does mean the components should be read as directions of variance rather than as latent factors with any distributional interpretation, and it justifies the use of Spearman rather than Pearson correlation throughout.

Table 3: Headline per-90 statistics by position group in the primary season, reported as mean with standard deviation in parentheses.

Statistic	DF	MF	FW
np_xg_p90	0.05 (0.04)	0.16 (0.12)	0.41 (0.17)
xa_p90	0.06 (0.06)	0.15 (0.10)	0.16 (0.11)
key_passes_p90	0.51 (0.44)	1.26 (0.63)	1.05 (0.54)
xg_chain_p90	0.35 (0.16)	0.50 (0.20)	0.63 (0.20)
xg_buildup_p90	0.30 (0.14)	0.28 (0.13)	0.18 (0.09)
shots_p90	0.54 (0.33)	1.54 (0.85)	2.53 (0.69)
shot_accuracy_pct	29.22 (20.60)	31.91 (13.77)	40.30 (9.65)
crosses_p90	1.22 (1.62)	2.26 (1.75)	1.40 (1.62)
interceptions_padj_p90	0.96 (0.37)	0.74 (0.44)	0.24 (0.17)
tackles_won_padj_p90	1.07 (0.48)	1.16 (0.54)	0.56 (0.34)
fouls_committed_padj_p90	0.87 (0.39)	1.31 (0.55)	1.28 (0.61)
offsides_p90	0.06 (0.07)	0.16 (0.17)	0.53 (0.33)

4.5 Statistics that change meaning by position

The correlation analysis produced the clearest substantive finding available from the reduced feature set. Correlations were computed within each position group and compared across groups, and the ten pairs whose correlation varies most between groups are reported in Table 4 and shown in Figure 3. The largest spread between position groups reaches 0.87, against a median spread across all pairs of 0.38.

Every one of the ten most divergent pairs changes sign between position groups. The relationship between shots on target and crosses is positive for midfielders and negative for forwards, which is to say that a midfielder who crosses often is generally a more active attacker while a forward who crosses often is generally not the player getting on the end of chances. The relationship between key passes and tackles won is positive for defenders and negative for midfielders, distinguishing the full back whose defensive work and creative output rise together from the midfield division of labour between a destroyer and a creator.

The practical implication is that a single pooled correlation matrix, of the kind that appears routinely in public football analysis, averages over these sign reversals and reports a number that describes no actual population of players. It also supplies the central justification for the position stratified analysis that follows, since a global embedding fitted across all outfield players is fitted across relationships that point in opposite directions.

4.6 Comparing embeddings

Principal component analysis, UMAP and t-SNE were fitted to the same standardised features and compared in Figure 4. The selected UMAP configuration used 10 neighbours at a minimum distance of 0, chosen by trustworthiness over the twelve configuration grid reported in Appendix C.

The three methods agree on the dominant structure and differ in how they present it. Trustworthiness is 0.855 for the linear projection, 0.932 for UMAP and 0.944 for t-SNE, while continuity is close to identical across the three. Nearest neighbour position purity at ten neighbours is 0.649 for the linear projection and 0.699 for UMAP, so roughly two thirds to three quarters of a player’s closest statistical neighbours share his listed position group. That figure is informative in both directions: it confirms that the surviving features carry real positional signal, and it establishes that a substantial minority of every player’s nearest neighbours are listed at a different position,

Distribution by position group, 2024-25 (white tick marks the median)

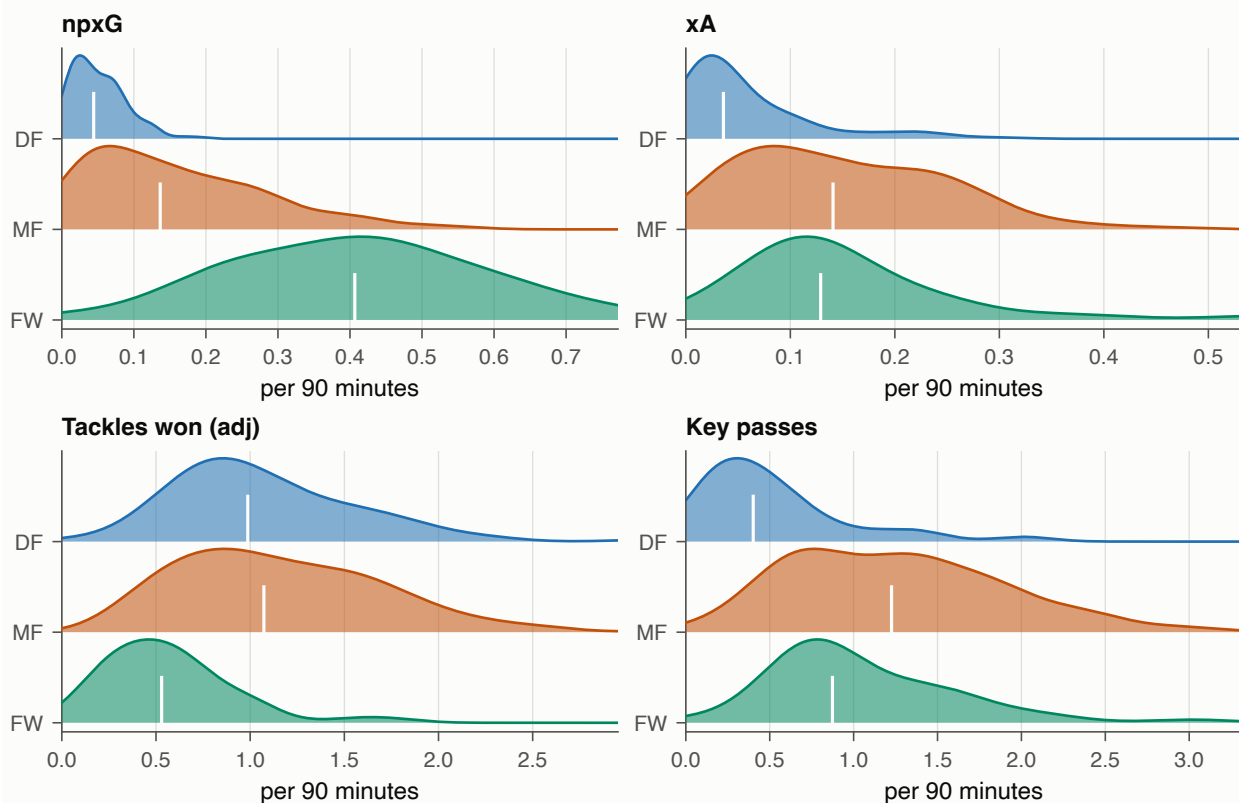


Figure 2: Distributions of four representative per-90 statistics across position groups in the primary season. The separation between groups is clear for shot volume and for defensive actions, and much weaker for creation.

which is the space in which data-driven roles can differ from listed ones.

All three embeddings display a continuous gradient from defenders through midfielders to forwards rather than separated groups. That gradient, and not a set of islands, is the honest description of the geometry, and it conditions how the clustering that follows should be read.

4.7 The third dimension

Two components are a convenient way to draw the space and a poor way to describe it. The first two carry shooting volume and wide creation and together account for 0.521 of the variance. The third carries duels and discipline, loading on fouls committed, cards and tackles won, and it is orthogonal to both of the others, so a planar projection does not compress it, it deletes it. Retaining it raises captured variance to 0.634. Figure 5 shows the outfield space in all three, drawn from two viewpoints so that structure can be told apart from the accident of a single projection.

Adding the dimension does not change the headline geometry. The defender to forward gradient remains continuous rather than resolving into islands, which is the same conclusion the planar embeddings reached and is now established against a projection that had the opportunity to falsify it. Figure 6 places the selected partition in the same space: the boundary cuts across the shooting axis and is close to indifferent to duels and discipline, which is consistent with a split that is substantially about whether a player shoots.

Table 4: The ten feature pairs whose Spearman correlation differs most across position groups. Every pair changes sign between at least two groups.

Feature A	Feature B	DF	MF	FW	Spread
Shots on target	Crosses	-0.01	0.43	-0.44	0.87
Crosses	Shot accuracy %	-0.22	0.26	-0.57	0.82
Crosses	Interceptions (adj)	0.21	-0.50	0.30	0.81
npG	Crosses	-0.15	0.36	-0.43	0.79
Key passes	Tackles won (adj)	0.47	-0.32	0.26	0.78
Crosses	Tackles won (adj)	0.51	-0.25	0.48	0.76
xA	Shots on target	0.17	0.56	-0.19	0.75
Shots	Interceptions (adj)	0.22	-0.50	-0.04	0.72
xA	Tackles won (adj)	0.39	-0.30	0.15	0.69
xA	Interceptions (adj)	0.19	-0.50	0.16	0.69

The third dimension does change the club picture, and this is the one place where it produces a finding the planar analysis could not. Club centroids correlate with final league position at -0.861 on the first component and -0.695 on the second, but only 0.209 on the third. Shooting volume and wide creation both track league position; the physical dimension of duels, fouls and cards essentially does not. A club does not finish higher by fouling more or less, which is unsurprising stated plainly and is worth stating because the duelling axis is a full component of player variation that carries almost no information about team success. Figure 7 shows the clubs, labelled at the extremes of the table.

Finally, the three-dimensional view settles a structural question raised in Section 4.10 by measurement rather than assertion. Drawing each archetype centroid as a direction from the origin, the two centroids within a position group have a cosine of -1 for defenders, -1 for midfielders and -1 for forwards. They are exactly anti-parallel, which is forced by standardising within group before a two-way split. Figure 8 shows the consequence: six archetypes occupy three lines through the origin, not six independent directions.

4.8 How many roles the data supports

Applying the consensus rule declared in Section 3 to the four selection criteria gives $k = 2$ for the global clustering and the same value within each of the three position groups. The full criterion curves appear in Figure 9 and Appendix D. This is the second principal result of the paper. The surviving public feature set supports a binary division within each position group and nothing finer. Requesting more clusters produces partitions that the criteria do not support and that resampling does not reproduce.

Stability separates the position groups sharply. Across 200 bootstrap resamples the mean adjusted Rand index against the reference partition is 0.914 globally and 0.815 for midfielders, both of which describe a partition that reappears almost unchanged under resampling. For defenders the mean falls to 0.607 with a standard deviation of 0.216, and for forwards to 0.602 with a standard deviation of 0.281. The minimum adjusted Rand index observed across forward resamples is -0.065, which is to say that in the least favourable resample the partition recovered is no better than chance. Forwards number only 49 players above the minutes threshold, so this is partly a sample size problem, but it is a real limit on what can be claimed. The consensus co-assignment structure is shown in Figure 10a.

The relationship between discovered clusters and listed positions is weak. The global partition

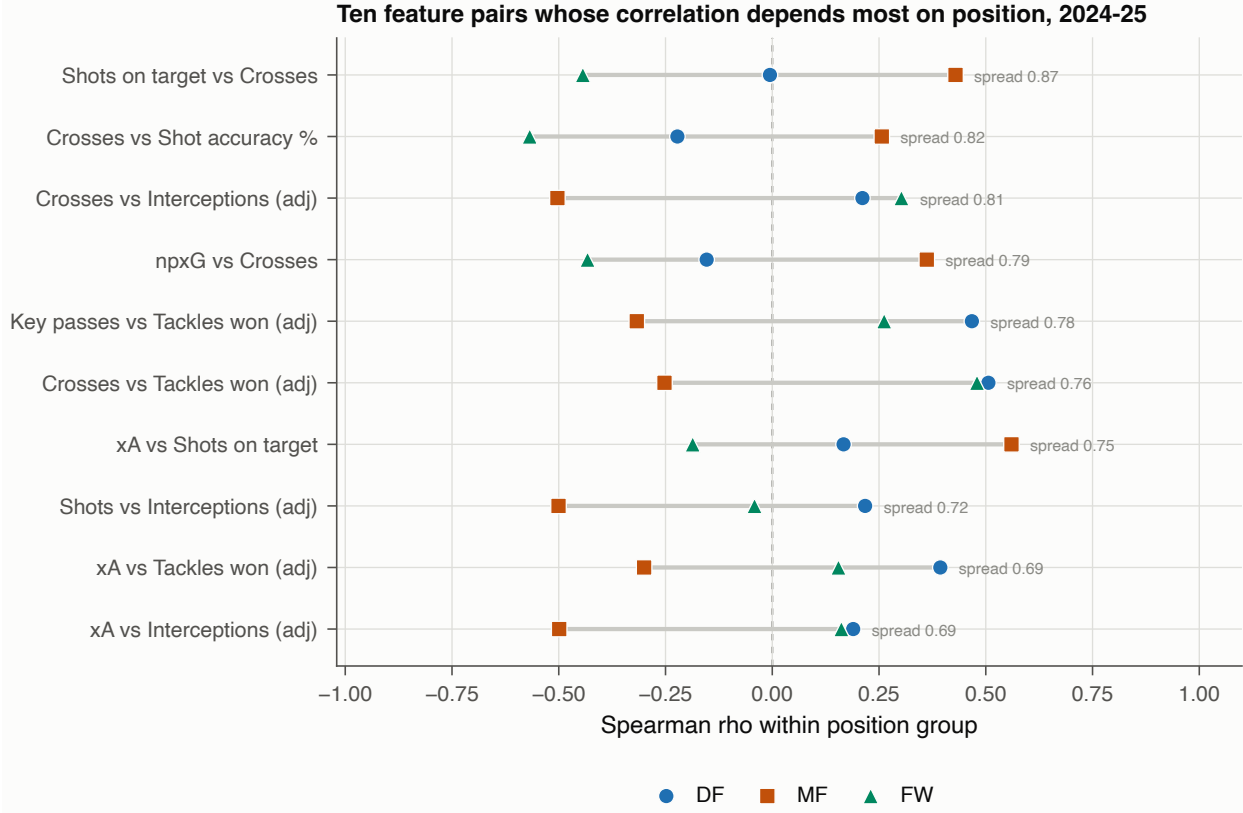


Figure 3: The ten feature pairs whose within-group correlation differs most across position groups in the primary season. Every pair changes sign between at least two groups.

scores an adjusted Rand index of 0.226 against the four-way position group with a cluster purity of 0.579, so the clusters are not a rediscovery of the positions that the source already records. Figure 10b shows where the two disagree. Read together with the embedding geometry, the interpretation is that the dominant axis in the surviving data is not defender against forward but attacking involvement against defensive involvement, which cuts across the listed groups and places a ball-playing full back nearer an ordinary midfielder than a defensive counterpart at his own listed position.

4.9 Goalkeepers

The goalkeeper analysis is the clearest illustration of what the withdrawal cost. 23 of the 26 statistical columns in the advanced goalkeeping table are empty, and the 3 that survive are duplicated in the basic table, so the advanced table contributes nothing whatsoever. Of the five composite axes the study set out to build, 2 can be constructed and 4 cannot. No proxy is substituted for the missing four, because no surviving column measures sweeping, distribution or cross handling even approximately.

With post-shot expected goals unavailable, shot stopping is estimated by regressing saves on shots on target faced and treating the residual as saves above volume. The regression returns a coefficient of determination of 0.985, which is high almost by construction, since every shot on target is either saved or conceded. The informative quantity is the residual spread, a standard error of 4.69 saves across the season. This index adjusts for how much work a goalkeeper receives but not

Two-dimensional embeddings of the same 18 features, 2024-25
T trustworthiness, C continuity, purity at k=10

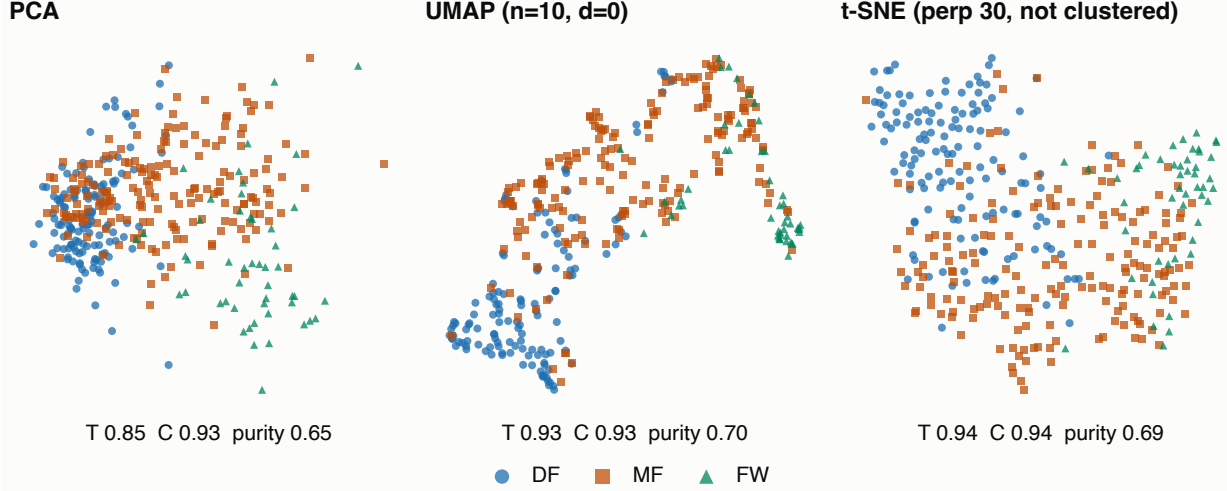


Figure 4: Two-dimensional embeddings of the same feature set, coloured by listed position group. The defender to forward gradient is continuous in all three, and t-SNE is shown for visual contrast only and is not used for clustering.

for how hard that work is, which is precisely what post-shot expected goals provided, so it should be read as a weak substitute rather than an equivalent.

Save percentage shows no relationship with workload. The correlation is -0.247 in the primary season at a p-value of 0.166, and it changes sign to 0.056 in the replication season. A relationship whose sign is not stable across two seasons is the strongest available form of a null result.

Goalkeeper clustering does not produce meaningful structure, and we report that rather than presenting a partition. The same consensus rule selects 2 clusters in the primary season at a silhouette of 0.495 but 4 clusters in the replication season at 0.576, so the cluster count does not replicate. The smallest replication cluster contains 3 members, below the floor fixed before fitting. Most decisively, cluster membership explains more variance in situational quantities than in goalkeeping ones: the proportion of variance explained is 0.72 for team points per match against 0.25 for save percentage. The partition recovers the strength of the team in front of the goalkeeper, not the goalkeeper. No archetype names are assigned.

Figure 13 ranks eligible goalkeepers on the saves above volume index, and Figure 14a places them on the two axes that survive. The distribution of both composites is given in Figure 27b, and the correlation structure of the surviving goalkeeper features in Figure 27a. The partition itself is shown in Figure 14b, presented for completeness rather than as a finding, for the reasons given above.

4.10 Archetypes

Because the selection rule returned two clusters in each position group, the study describes six outfield archetypes rather than the richer taxonomy the design anticipated. Their centroid profiles are shown in Figure 15 and their radar comparisons in Figure 16. Names were assigned only after inspecting the distinguishing features and the member players, and the justification for each is reproduced from the generated record rather than composed for the paper.

Among defenders the split separates 98 players with no positive attacking signature at all from

Outfield players in three principal components, 2024-25

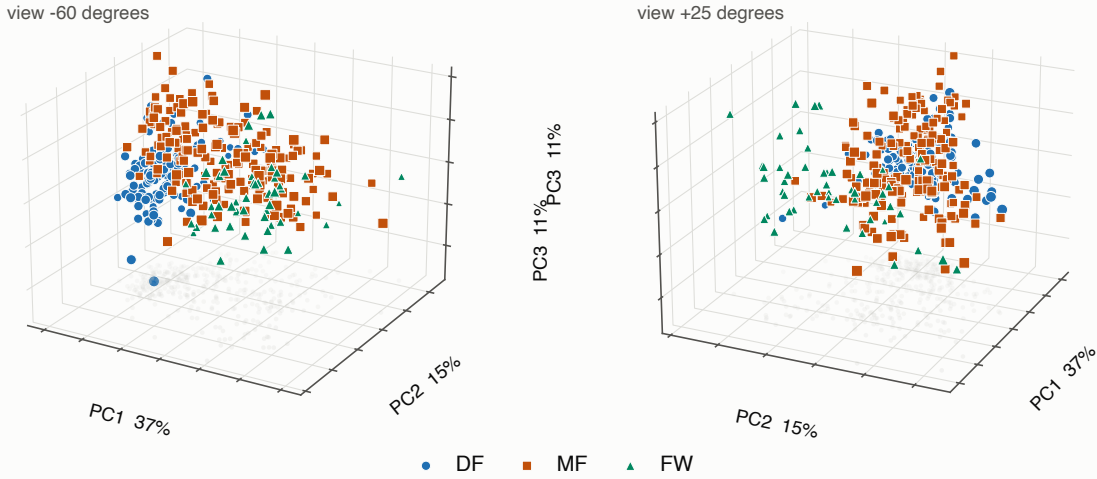


Figure 5: Outfield players in the first three principal components, from two viewpoints. Grey points on the floor are the vertical projection of each player, supplying a depth cue that a static rendering otherwise lacks.

38 whose key passes, assists, expected assists and crosses all sit around one standard deviation above the defensive mean. The first group is named for what it does not do, because every one of its eight distinguishing features is negative and its largest positive deviation is negligible. The second corresponds closely to full backs and wing backs. Among midfielders the division is the cleanest in the study, separating 100 players whose only positive deviations are defensive or disciplinary from 81 whose shooting and creation measures are all elevated. Among forwards the split distinguishes 30 central players who shoot from 19 wide players who cross and create.

One structural property must be stated because it limits how much the six profiles can be read as independent. Features are standardised within position group before clustering, so with two clusters the two centroids in each group are necessarily anti-parallel: their size-weighted sum is zero. Each position group therefore yields two ends of a single axis rather than two independently shaped profiles, which is why the two members of each pair mirror one another so exactly. The axis is real and the ordering along it is informative, but the six archetypes represent three axes, not six independent types.

Outlier analysis is reported in Figure 22a. The isolation forest and the Mahalanobis distance agree closely, which is reassuring given that one is a partition method and the other a distance method.

4.11 What a supervised model sees

The clustering results leave one question open: are the surviving features too weak to carry positional information at all, or do they carry it while the clustering chooses to partition along a different axis? A supervised model settles it. Predicting the listed position group from the same 18 features, gradient boosting reaches an accuracy of 0.803 and a macro F1 of 0.769, and logistic regression reaches 0.811 and 0.786, against a majority class baseline of 0.494 and 0.221. The features clearly do carry position, so the weak agreement between clusters and positions reflects the clustering

The global partition in three components, crosses mark centroids, 2024-25

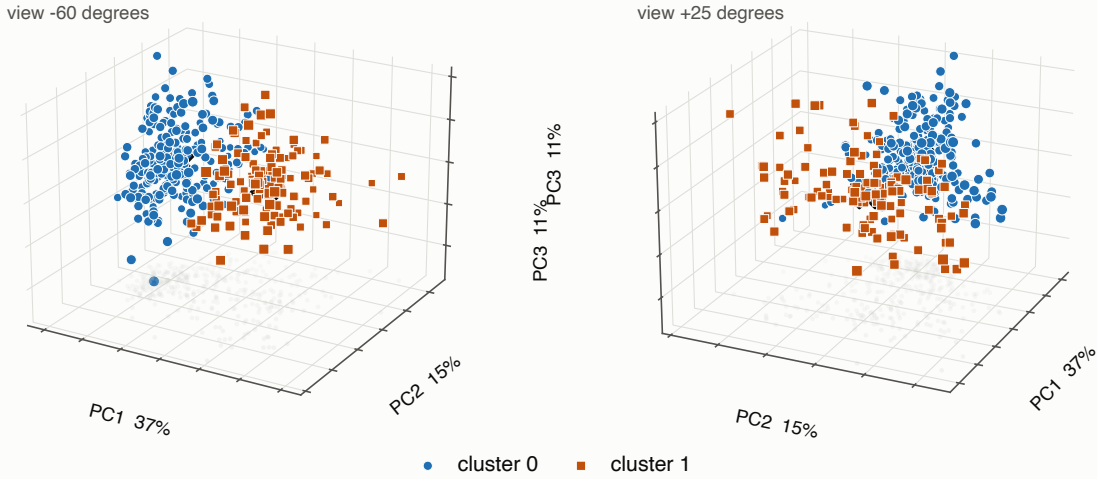


Figure 6: The selected global partition in three components, with cluster centroids marked by crosses. The split runs along the shooting axis and is close to indifferent to the third component.

finding a different and equally real axis, not an empty feature set. The confusion is almost entirely between adjacent groups, as Figure 17a shows, and forwards are the weakest class.

The SHAP decomposition in Figures 23 and 17b carries a finding that follows directly from the withdrawal. With possession and passing volume gone, positions are identified largely by absence. Defenders are recognised by low shooting and low key passes, forwards by low interceptions, and midfielders by an intermediate profile on both. A classifier of this kind, trained before January 2026, would have identified a defender by what he did with the ball. It now identifies him by what he does not do.

Players the classifier places in the wrong group overlap significantly with the players the clustering flagged as hybrids. The overlap rate is 0.623 for logistic regression at an enrichment of 1.48 over independence, and 0.556 at an enrichment of 1.32 for gradient boosting. The enrichment is modest because the hybrid flag is itself broad, covering 154 of 366 players, so this is corroboration rather than a strong result.

The finishing above role regression illustrates a hazard worth recording. Predicting non-penalty expected goals per ninety from non-shooting behaviour returns a cross validated coefficient of determination of 0.531. An earlier specification that also included xGChain returned 0.892, which looks far better and is in fact circular: Understat credits xGChain for possessions the player finished himself, so the difference between xGChain and xGBuildup contains the target and correlates with it at 0.904. The apparently strong model was reconstructing the outcome rather than predicting it. Even at the honest value, roughly half the variance is unexplained, so the residual leaderboard in Figure 22b is presented as an illustration and not as a measurement of finishing ability.

4.12 Team signatures

Aggregating players into clubs produces the one result in this study that is both strong and stable. The minutes-weighted position of a club's outfield players on the first principal component correlates with final league position at a Spearman coefficient of -0.771 in the primary season and -0.783 in the

**Clubs as minutes-weighted centroids, darker finished higher, 2024-25.
Only the top and bottom 5 are labelled**

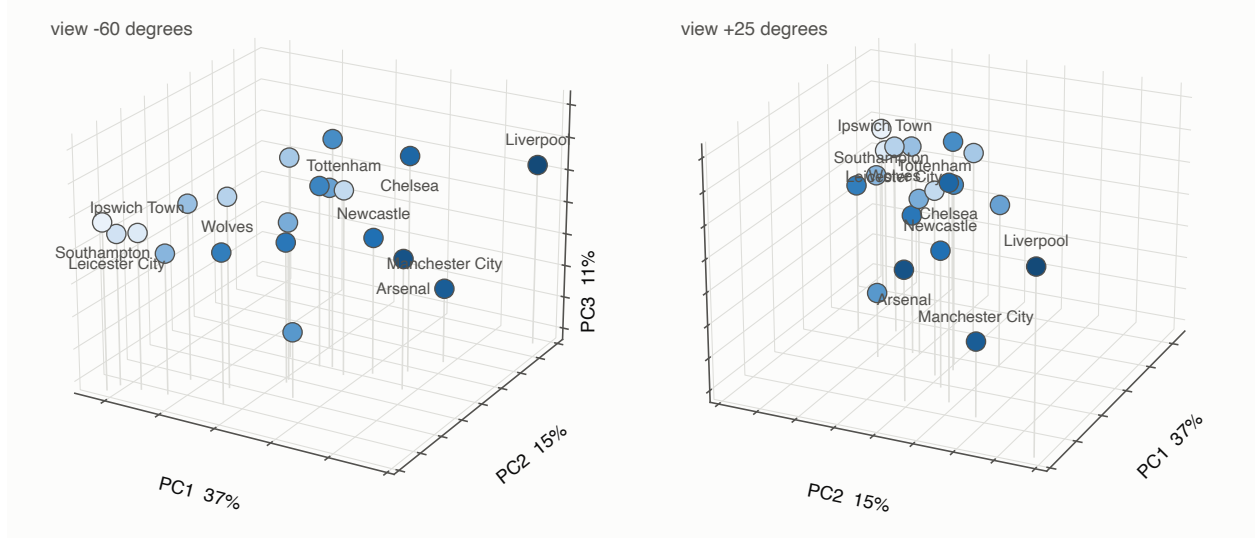


Figure 7: Clubs as minutes-weighted centroids in three components, shaded by final league position with darker shading for a higher finish. Vertical lines to the floor give the depth cue, and only the top and bottom of the table are labelled.

replication season, which is close to an exact reproduction. Since the first component is essentially attacking involvement, this says that clubs whose players collectively shoot and create more finish higher, which is unsurprising in itself, but the near identical coefficient across two independent seasons shows the team level summary is measuring something stable even where the player level partitions are not.

Hierarchical clustering on archetype composition groups the 20 clubs into 3 style families, shown in Figure 18a, with composition by club in Figure 18b. Figure 19 places each club's players in the shared embedding against the rest of the league, which makes squad shape legible in a way that a composition bar cannot: clubs differ visibly in how far their defenders sit toward the midfield region.

4.13 Similarity, replication and noise

4.13.1 Player similarity

Nearest neighbour search within position group recovers football sense for players whose roles are defined by the surviving statistics, and fails informatively for players whose roles are not. Full backs return full backs, penalty box forwards return penalty box forwards, and centre backs return centre backs. The instructive failures are the midfielders. A holding midfielder's nearest neighbours are creative attacking midfielders, because with no tackles by pitch third, no blocks, no clearances, no pressures and no passing volume, the only things left that distinguish him are his set piece deliveries and key passes. The metric finds the players who share that signature and misses the entire defensive half of his job. An inverted full back returns three centre backs for the same reason, since the role is defined by possession behaviour that no longer exists in the data. Similarity scores among centre backs are systematically the lowest in the table, which is the same feature poverty appearing as an inability to tell defenders apart at all. Figure 20a shows the structure.

Archetype centroid directions: each position group is one axis, not two

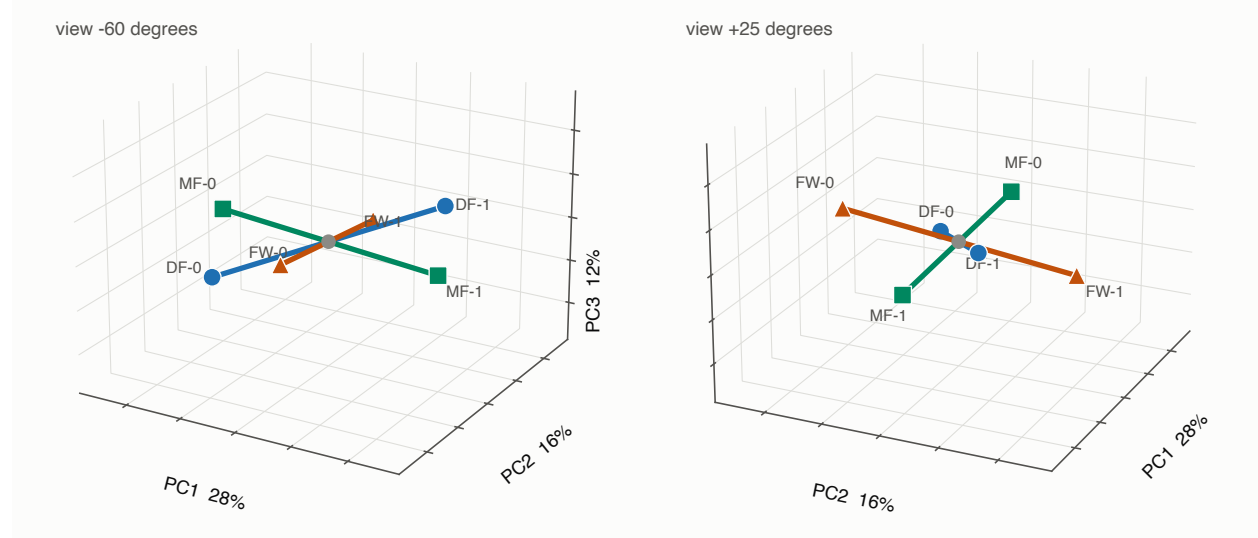


Figure 8: Archetype centroid directions, drawn as unit vectors so the figure shows orientation rather than magnitude. Each position group forms a single line through the origin, so the six archetypes describe three axes.

4.13.2 Cross-season replication

242 players are eligible in both seasons. The declared rule independently selects the same number of clusters in every scope in both seasons, so the structure replicates. Membership replicates only partly. The adjusted Rand index between seasons is 0.598 globally, with 27 players changing cluster, and within position group it falls to 0.526 for defenders, 0.381 for midfielders and 0.453 for forwards. The within group figures are worse than the bootstrap stability of Section 4.8, which is what would be expected if the within group splits are finer than the data can reliably support. Figure 20b shows the transitions.

4.13.3 Sensitivity to sampling noise

Empirical Bayes shrinkage toward the position group mean, with the degree of shrinkage set by minutes played, provides the sharpest test of whether the partitions describe playing roles or sampling noise. Shrinkage is substantial rather than cosmetic, and the results divide cleanly.

The global partition survives. Only 5 of 366 players change cluster, the adjusted Rand index between the original and shrunk partitions is 0.945, the selected number of clusters is unchanged, and the rate at which players move falls monotonically with minutes played, exactly as the noise account predicts.

The within group partitions fare worse, and they do not fare equally. For defenders the adjusted Rand index falls to 0.453, with 21 of 136 players moving, which is a partition largely dissolved by removing sampling noise. Midfielders retain 0.770 and forwards 0.626, both of which survive in part rather than collapsing. The forward figure is notably higher than it was before the possession correction of Section 4.2 was applied, which is itself informative: some of what previously looked like noise in the forward partition was exposure to opponent possession, and removing it left a more reproducible split. The mechanism is visible in the estimated variance components: several features

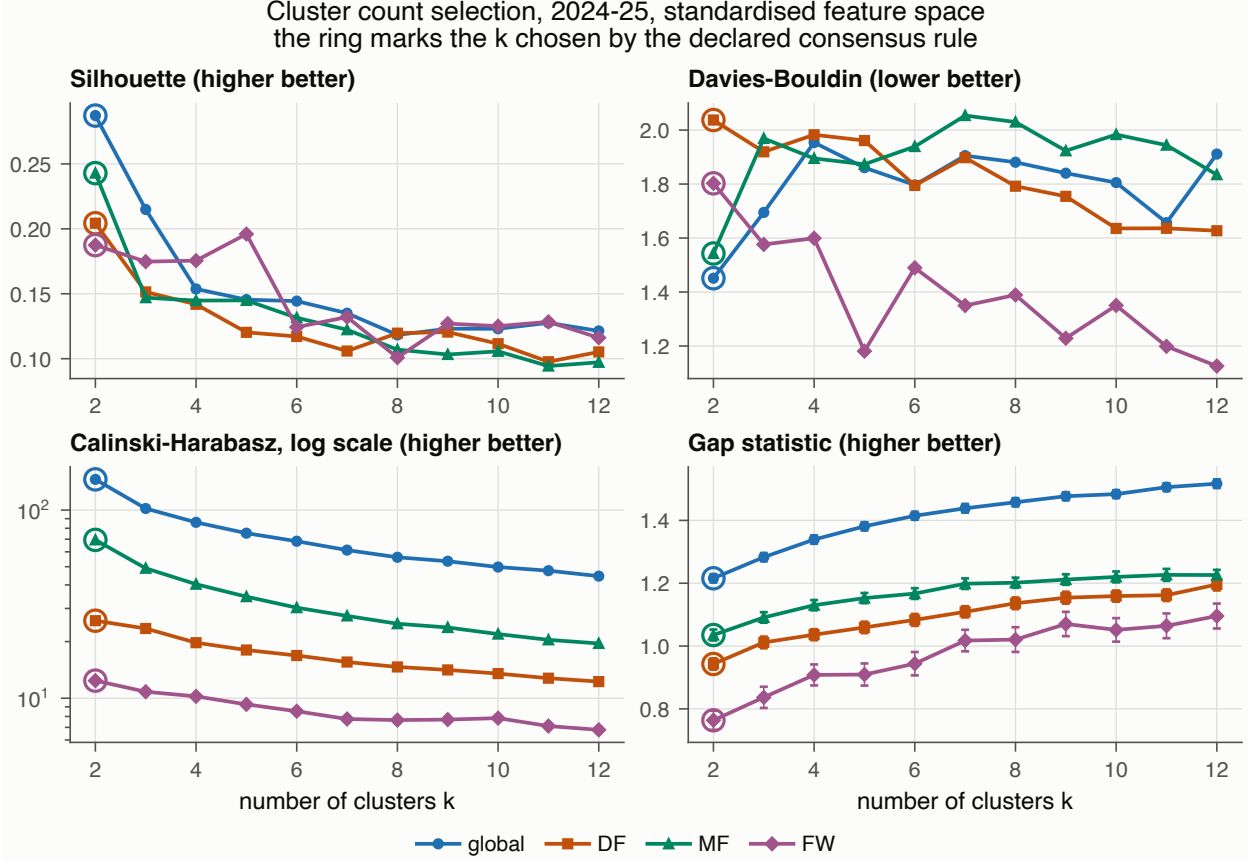
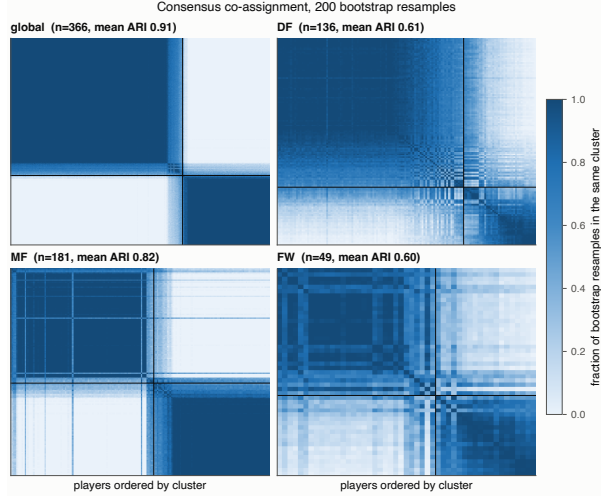


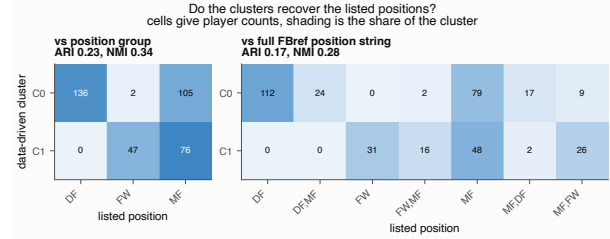
Figure 9: Cluster selection criteria across candidate cluster counts. The consensus rule was fixed before fitting and selects the same value in every scope.

are assigned an estimated between-player variance of zero within a position group, meaning the model attributes their entire observed spread to sampling noise and collapses them to the group mean, removing them from the clustering altogether. Shot accuracy retains very little reliability, because its true spread between players is small relative to the noise in estimating it from a season of shots.

Read alongside the bootstrap and replication results, the three tests do not deliver a single uniform verdict, and reporting them as though they did would overstate the case. Defenders fail all three: bootstrap stability of 0.607, cross-season agreement of 0.526, and near dissolution under shrinkage. Forwards fail two of the three, with a bootstrap minimum of -0.065 across resamples and cross-season agreement of 0.453, while partly surviving shrinkage. Midfielders pass all three. The conclusion that survives every test is therefore narrower than a single summary sentence would suggest: the global division between attacking and defensive involvement is a real property of players, the midfield split is trustworthy, and the defender split should not be used to characterise an individual at all. Figures 21a and 21b show the effect.



(a) Bootstrap co-assignment.



(b) Clusters against listed positions.

Figure 10: Cluster stability and the correspondence with listed positions.

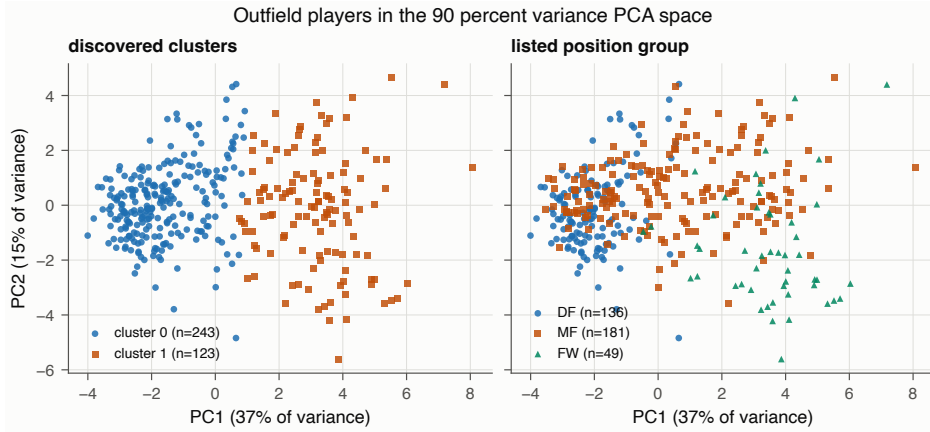


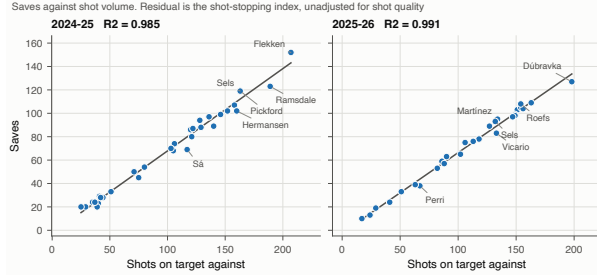
Figure 11: The selected partition shown in the principal component space.

5 Discussion

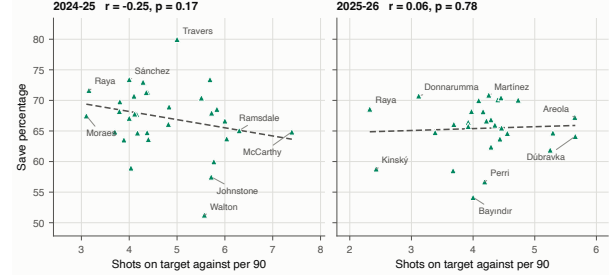
5.1 What the archetypes mean, and where they break

The six outfield archetypes describe three axes rather than six independent types, since standardising within position group before a two-way split forces the two centroids in each group to be mirror images. Each axis orders players from low to high attacking involvement within their listed group, and in every group the high involvement end is distinguished by crossing and chance creation while the low involvement end is distinguished by defensive actions or, in the case of defenders, by the absence of anything positive at all. That an ordering exists is a real result. That it is a single ordering rather than a taxonomy is the limit the data now imposes.

Bootstrap resampling, replication on an independent season and shrinkage toward the position mean test three different failure modes, and they do not return a single verdict. The midfield split passes all three. The defender split fails all three. The forward split fails two, reaching an adjusted Rand index below zero in the least favourable bootstrap resample, meaning the partition recovered



(a) Saves against shots on target faced.



(b) Save percentage against workload.

Figure 12: Goalkeeper shot stopping and workload, the only two dimensions the surviving data supports.

there was worse than a random relabelling, while partly surviving shrinkage once defensive counts are corrected for possession. A reader who wants one sentence of practical guidance should take this one: use the global attacking involvement axis, treat the midfield split as usable, treat the forward split as suggestive at best given only 49 eligible players, and do not characterise an individual defender by his within-group cluster.

The goalkeeper analysis deserves separate emphasis because it failed for a reason worth recording. The partition that the selection rule returned explains more variance in team points per match than in save percentage. In other words the goalkeeper clusters recover the quality of the team in front of the goalkeeper rather than the goalkeeper himself. With post-shot expected goals removed, almost every surviving goalkeeper statistic is a function of how much and what quality of shooting the defence permits, so this is close to inevitable. Reporting the partition without that check would have produced plausible looking goalkeeper archetypes that were actually a league table in disguise.

5.2 Listed positions and data-driven roles

The clusters do not recover listed positions, and the interesting part is where they disagree. The global division places every defender and a majority of midfielders on one side, with the remaining midfielders and every forward on the other, so it cuts the midfield in half and leaves the defender and forward groups intact. The players it reassigns are recognisable: holding midfielders sit with the defenders, and attacking midfielders sit with the forwards. That the same players are also the ones a supervised classifier places in the wrong group is corroboration from a second method, although the enrichment over chance is modest because the hybrid flag is broad.

The similarity search makes the limitation concrete in a way the aggregate statistics do not. A holding midfielder’s nearest statistical neighbours are creative attacking midfielders, because the surviving features contain nothing that describes his actual job. Tackles by pitch third, blocks, clearances, pressures and passing volume are all gone, so the only distinctive things left in his profile are set piece deliveries and key passes, and the metric duly finds the players who share those. An inverted full back returns centre backs for the same reason. These are not failures of the method but measurements of what the data no longer contains, and they are the clearest illustration in the paper of what the withdrawal cost.

5.3 Limitations

Season aggregates discard context: a tackle count does not say where the tackles happened, and an expected goals total does not distinguish six poor chances from two good ones. The defensive counts

are possession adjusted, using an elasticity measured from the data rather than the conventional assumption of direct proportionality, which Section 4.2 shows would have overcorrected badly. That removes the exposure confound from those three features, but it does not make the feature set team-neutral overall: the attacking statistics still scale with a team’s own share of the ball, and the first principal component remains strongly related to team strength. Correcting both directions would require an instrument this data cannot supply, so the residual confound is acknowledged rather than solved. Per-90 rates remain noisy above the minutes threshold, which is why the shrinkage analysis matters and why its within-group verdict should be believed. Merging a second expected goals provider introduces a model whose assumptions differ from the one that produced the original values, and although we never mix the two sources for the same quantity, the creation block and the shooting block now come from different modelling traditions. Roughly three percent of records failed to match across the two sources and carry missing values for the Understat block. The study covers one league and two seasons. Finally, the clusters are descriptive and carry no causal content: an archetype label records how a player’s season looked, not why, and certainly not how he would perform in a different team.

5.4 Applications

For scouting and recruitment the honest use of this work is narrow but real. The global involvement axis and the club level signatures are stable enough to support questions about squad composition, such as whether a club’s minutes are concentrated in one part of the distribution, and the twenty panel comparison makes that legible at a glance. The similarity search is usable for forwards and full backs, where the surviving features capture most of what defines the role, and should not be used for central midfielders or centre backs, where they do not. For tactical analysis the correlation divergence result is the most directly applicable: any analysis that pools across positions when computing relationships between statistics is averaging over sign reversals.

5.5 Future work

The obvious extension is data. Event and tracking data would restore everything the withdrawal removed and more, at the cost of the reproducibility that a public source provides. Within public data, match level statistics remain available in a reduced form and would support the role drift analysis that was scoped out of this study, tracking whether a player’s profile moves between archetypes across a season rather than assuming a season is homogeneous. Extending to other leagues would test whether the two-cluster result is a property of the reduced feature set or of the Premier League specifically, and we expect the former. Symmetric adjustment of the attacking statistics was tested and deliberately not adopted, for the reasons given in Section 4.3. A reader who disagrees with that judgement would find the next step in modelling attacking exposure properly, since a player’s opportunity depends on where his team uses the ball and not only on how much of it it has, which a single elasticity cannot capture. Finally, the empirical Bayes machinery developed here is more broadly useful than the single use it is put to: any per-90 ranking computed over partial seasons is subject to the same problem, and most published ones do not correct for it.

6 Conclusion

The withdrawal of commercially licensed statistics from the largest free football data source in January 2026 removed most of the vocabulary that public role analysis had come to depend on. This paper measured what remains. Merging the surviving statistics with an independent expected goals

source yields 18 usable features, and on those features a role discovery pipeline recovers a single robust division between attacking and defensive involvement, which cuts across listed positions rather than reproducing them, and little else that survives scrutiny.

The negative results are the substantive ones. The selection rule returns two clusters in every scope. Density based clustering, which is free to conclude that no clusters exist, concludes exactly that within every position group. Bootstrap resampling, cross-season replication and empirical Bayes shrinkage together show that the defender partition is substantially a description of sampling noise, and that the forward partition is not reproducible enough to characterise individuals. The goalkeeper partition recovers team strength rather than goalkeeping. Reporting these as findings rather than suppressing them is the point: a study of this kind run without the stability checks would have produced six confidently named archetypes and a goalkeeper taxonomy, and most of it would have been wrong.

Two results are strong enough to carry forward. Correlations between statistics reverse sign across position groups often enough that pooled correlation analysis of football data should be considered unsafe by default. Club level summaries built from the same features correlate with final league position at almost identical strength across two independent seasons, so aggregation to the club recovers stability that individual players do not have.

The broader lesson concerns dependence. An entire layer of public football analysis rested on a single source holding a single commercial licence, and when that licence ended the layer went with it. The work here is reproducible from a public pipeline, which is what made it possible to measure the loss precisely rather than estimate it, and that reproducibility is worth more now than it was before January 2026.

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Table 5: The outfield feature set. Counting statistics are per ninety minutes. The two rate features are used as published.

Feature	Source	Definition
np_xg_p90	Understat	Non-penalty expected goals
xa_p90	Understat	Expected assists
key_passes_p90	Understat	Passes leading directly to a shot
xg_chain_p90	Understat	Expected goals of possessions the player was involved in
xg_buildup_p90	Understat	As xGChain, excluding shots and key passes
shots_p90	FBref	Shots taken, excluding penalties
shots_on_target_p90	FBref	Shots on target
goals_non_penalty_p90	FBref	Goals excluding penalties
assists_p90	FBref	Assists
crosses_p90	FBref	Crosses attempted
interceptions_padj_p90	FBref	Interceptions, possession adjusted
tackles_won_padj_p90	FBref	Tackles won, possession adjusted
fouls_committed_padj_p90	FBref	Fouls committed, possession adjusted
fouls_drawn_p90	FBref	Fouls drawn
offsides_p90	FBref	Times caught offside
cards_yellow_p90	FBref	Yellow cards
shot_accuracy_pct	FBref	Shots on target as a share of shots, a rate
goals_per_shot	FBref	Goals per shot, a rate

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A Feature dictionary

Table 5 lists every feature in the outfield set, its source, and its definition. Counting statistics are expressed per ninety minutes played. The two rate features are used as published rather than divided a second time. Each feature was confirmed to hold non-null values in both seasons before being admitted, which is the check that distinguishes a live column from one whose values were withdrawn.

B Minutes threshold sensitivity

The main analysis excludes outfield players below 450 minutes. That threshold balances two errors: a low threshold admits players whose per-90 rates are dominated by sampling noise, while a high threshold discards genuine squad roles such as the impact substitute. Table 6 repeats the entire selection and stability procedure at 270, 450 and 900 minutes, treating each threshold as an independent pool in which imputation and standardisation are recomputed and the declared selection rule is reapplied from scratch. Figure 24 plots the same quantities.

The threshold turns out to matter very little. The rule selects two clusters at every threshold, the silhouette moves only from 0.286 to 0.302, and bootstrap stability is between 0.924 and 0.947 throughout. More directly, the partition itself barely moves: measured on the players common to both pools, the adjusted Rand index against the partition at the threshold actually used is 0.967 at

Table 6: Sensitivity of the analysis to the minutes threshold in the primary season. Agreement is the adjusted Rand index against the partition at the threshold actually used, computed on the players common to both pools.

Minutes	Eligible	k	Silhouette	Bootstrap ARI	Agreement
270	403	2	0.286	0.930	0.967
450	366	2	0.286	0.924	baseline
900	296	2	0.302	0.947	0.986

Table 7: UMAP hyperparameter grid, ranked by trustworthiness. Selection was by trustworthiness alone and was decided before any embedding was inspected.

Neighbours	Minimum distance	Trustworthiness
10	0	0.9322
10	0.10	0.9316
15	0	0.9306
15	0.10	0.9303
30	0.10	0.9276
30	0	0.9257
10	0.50	0.9186
15	0.50	0.9175
50	0.10	0.9169
50	0	0.9156
30	0.50	0.9083
50	0.50	0.9045

270 minutes and 0.986 at 900. A reader who preferred a different threshold would reach the same conclusions.

C UMAP hyperparameter grid

Table 7 reports trustworthiness for all twelve configurations of the neighbour count and minimum distance. Selection was by trustworthiness alone, decided before any embedding was inspected visually.

D Cluster selection criteria in full

Table 8 gives the silhouette coefficient, Davies-Bouldin index, Calinski-Harabasz index and gap statistic for every candidate number of clusters, for the global analysis and within each position group. The consensus rule described in Section 3 is applied to these values.

Table 8: Cluster selection criteria for every candidate number of clusters, by scope. The consensus rule is applied to these values.

Scope	k	Silhouette	Davies-Bouldin	Calinski-Harabasz	Gap
global*	2	0.287	1.451	145.6	1.216

Scope	k	Silhouette	Davies-Bouldin	Calinski-Harabasz	Gap
global	3	0.215	1.695	101.9	1.283
global	4	0.154	1.954	86.1	1.339
global	5	0.145	1.861	75.4	1.381
global	6	0.144	1.797	68.3	1.415
global	7	0.135	1.905	61.3	1.438
global	8	0.118	1.881	56.1	1.458
global	9	0.123	1.840	53.5	1.477
global	10	0.123	1.805	49.8	1.484
global	11	0.128	1.657	47.6	1.506
global	12	0.121	1.911	44.6	1.517
DF*	2	0.204	2.037	25.8	0.943
DF	3	0.151	1.919	23.5	1.011
DF	4	0.142	1.983	19.7	1.036
DF	5	0.120	1.961	18.0	1.059
DF	6	0.117	1.795	16.9	1.083
DF	7	0.106	1.898	15.6	1.109
DF	8	0.120	1.792	14.7	1.136
DF	9	0.120	1.754	14.1	1.154
DF	10	0.112	1.636	13.5	1.159
DF	11	0.098	1.636	12.8	1.162
DF	12	0.105	1.627	12.3	1.196
MF*	2	0.243	1.543	69.4	1.035
MF	3	0.147	1.969	49.0	1.091
MF	4	0.145	1.895	40.4	1.130
MF	5	0.145	1.873	34.7	1.153
MF	6	0.132	1.939	30.3	1.167
MF	7	0.122	2.054	27.4	1.199
MF	8	0.107	2.030	24.9	1.202
MF	9	0.103	1.923	23.8	1.212
MF	10	0.106	1.983	21.9	1.220
MF	11	0.094	1.944	20.4	1.227
MF	12	0.097	1.835	19.5	1.226
FW*	2	0.187	1.803	12.4	0.764
FW	3	0.175	1.576	10.8	0.837
FW	4	0.176	1.599	10.2	0.908
FW	5	0.196	1.181	9.3	0.909
FW	6	0.124	1.490	8.5	0.944
FW	7	0.132	1.350	7.8	1.017
FW	8	0.101	1.389	7.7	1.021
FW	9	0.127	1.229	7.7	1.070
FW	10	0.125	1.350	7.8	1.051
FW	11	0.128	1.199	7.1	1.064
FW	12	0.116	1.126	6.8	1.096

E Cluster membership

Table 9 lists every eligible outfield player in the primary season with his club, listed position, assigned archetype and distance from his cluster centroid.

Table 9: Cluster membership for every eligible outfield player in the primary season.

Player	Club	Listed	Archetype	Distance
Aaron Wan-Bissaka	West Ham	DF	Crossing and chance creating defenders	3.09
Andy Robertson	Liverpool	DF	Crossing and chance creating defenders	4.59
Antonee Robinson	Fulham	DF	Crossing and chance creating defenders	4.28
Ashley Young	Everton	DF	Crossing and chance creating defenders	4.02
Ben White	Arsenal	DF	Crossing and chance creating defenders	2.68
Conor Bradley	Liverpool	DF	Crossing and chance creating defenders	6.54
Cristian Romero	Tottenham	DF	Crossing and chance creating defenders	4.57
Dean Huijsen	Bournemouth	DF	Crossing and chance creating defenders	3.65
Destiny Udogie	Tottenham	DF	Crossing and chance creating defenders	4.51
Djed Spence	Tottenham	DF	Crossing and chance creating defenders	2.51
Ian Maatsen	Aston Villa	DF	Crossing and chance creating defenders	5.18
Joško Gvardiol	Manchester City	DF	Crossing and chance creating defenders	4.51
Jurriën Timber	Arsenal	DF	Crossing and chance creating defenders	3.25
Keane Lewis-Potter	Brentford	DF	Crossing and chance creating defenders	3.78
Kenny Tete	Fulham	DF	Crossing and chance creating defenders	4.14
Kieran Trippier	Newcastle	DF	Crossing and chance creating defenders	4.42
Leif Davis	Ipswich Town	DF	Crossing and chance creating defenders	4.61
Lewis Hall	Newcastle	DF	Crossing and chance creating defenders	3.95
Lisandro Martínez	Manchester Utd	DF	Crossing and chance creating defenders	3.95
Lucas Digne	Aston Villa	DF	Crossing and chance creating defenders	4.50
Mads Roerslev	Brentford	DF	Crossing and chance creating defenders	4.53
Malo Gusto	Chelsea	DF	Crossing and chance creating defenders	2.37

Player	Club	Listed	Archetype	Distance
Marc Cucurella	Chelsea	DF	Crossing and chance creating de-fenders	3.73
Matheus Nunes	Manchester City	DF	Crossing and chance creating de-fenders	5.13
Mats Wieffer	Brighton	DF	Crossing and chance creating de-fenders	6.15
Michael Kayode	Brentford	DF	Crossing and chance creating de-fenders	4.18
Milos Kerkez	Bournemouth	DF	Crossing and chance creating de-fenders	2.82
Neco Williams	Nottingham	DF	Crossing and chance creating de-fenders	3.49
Nico O'Reilly	Manchester City	DF	Crossing and chance creating de-fenders	7.55
Oleksandr Zinchenko	Arsenal	DF	Crossing and chance creating de-fenders	5.56
Pedro Porro	Tottenham	DF	Crossing and chance creating de-fenders	4.60
Pervis Estupiñán	Brighton	DF	Crossing and chance creating de-fenders	2.76
Reece James	Chelsea	DF	Crossing and chance creating de-fenders	2.78
Riccardo Calafiori	Arsenal	DF	Crossing and chance creating de-fenders	5.54
Ryan Manning	Southampton	DF	Crossing and chance creating de-fenders	5.20
Tariq Lamptey	Brighton	DF	Crossing and chance creating de-fenders	5.15
Trent Alexander-Arnold	Liverpool	DF	Crossing and chance creating de-fenders	6.44
Trevoh Chalobah	Crystal Palace / Chelsea	DF	Crossing and chance creating de-fenders	3.86
Aaron Cresswell	West Ham	DF	Low attacking involvement de-fenders	4.53
Abdukodir Khusanov	Manchester City	DF	Low attacking involvement de-fenders	3.38
Adam Smith	Bournemouth	DF	Low attacking involvement de-fenders	3.13
Adam Webster	Brighton	DF	Low attacking involvement de-fenders	4.13
Archie Gray	Tottenham	DF	Low attacking involvement de-fenders	4.44
Axel Disasi	Aston Villa / Chelsea	DF	Low attacking involvement de-fenders	3.55
Axel Tuanzebe	Ipswich Town	DF	Low attacking involvement de-fenders	2.69
Ben Davies	Tottenham	DF	Low attacking involvement de-fenders	2.29
Caleb Okoli	Leicester City	DF	Low attacking involvement de-fenders	3.05
Calvin Bassey	Fulham	DF	Low attacking involvement de-fenders	1.45

Player	Club	Listed	Archetype	Distance
Cameron Burgess	Ipswich Town	DF	Low attacking involvement de-fenders	3.22
Chris Richards	Crystal Palace	DF	Low attacking involvement de-fenders	2.65
Conor Coady	Leicester City	DF	Low attacking involvement de-fenders	5.36
Craig Dawson	Wolves	DF	Low attacking involvement de-fenders	2.87
Dan Burn	Newcastle	DF	Low attacking involvement de-fenders	2.06
Dara O'Shea	Ipswich Town	DF	Low attacking involvement de-fenders	2.10
Diego Carlos	Aston Villa	DF	Low attacking involvement de-fenders	3.36
Emerson Palmieri	West Ham	DF	Low attacking involvement de-fenders	3.18
Emmanuel Agbadou	Wolves	DF	Low attacking involvement de-fenders	1.68
Ethan Pinnock	Brentford	DF	Low attacking involvement de-fenders	3.21
Ezri Konsa	Aston Villa	DF	Low attacking involvement de-fenders	3.18
Fabian Schär	Newcastle	DF	Low attacking involvement de-fenders	4.54
Gabriel Magalhães	Arsenal	DF	Low attacking involvement de-fenders	4.17
Harry Maguire	Manchester Utd	DF	Low attacking involvement de-fenders	4.80
Ibrahima Konaté	Liverpool	DF	Low attacking involvement de-fenders	3.34
Igor	Brighton	DF	Low attacking involvement de-fenders	3.80
Illia Zabarnyi	Bournemouth	DF	Low attacking involvement de-fenders	2.39
Issa Diop	Fulham	DF	Low attacking involvement de-fenders	1.92
Jack Stephens	Southampton	DF	Low attacking involvement de-fenders	2.43
Jacob Greaves	Ipswich Town	DF	Low attacking involvement de-fenders	1.92
Jake O'Brien	Everton	DF	Low attacking involvement de-fenders	4.00
Jakub Kiwior	Arsenal	DF	Low attacking involvement de-fenders	3.05
James Bree	Southampton	DF	Low attacking involvement de-fenders	3.61
James Justin	Leicester City	DF	Low attacking involvement de-fenders	2.82
James Tarkowski	Everton	DF	Low attacking involvement de-fenders	2.08
Jan Bednarek	Southampton	DF	Low attacking involvement de-fenders	4.44

Player	Club	Listed	Archetype	Distance
Jan Paul van Hecke	Brighton	DF	Low attacking involvement de-fenders	1.75
Jannik Vestergaard	Leicester City	DF	Low attacking involvement de-fenders	2.96
Jarell Quansah	Liverpool	DF	Low attacking involvement de-fenders	4.41
Jarrad Branthwaite	Everton	DF	Low attacking involvement de-fenders	2.12
Jean-Clair Todibo	West Ham	DF	Low attacking involvement de-fenders	2.98
Joachim Andersen	Fulham / Crystal Palace	DF	Low attacking involvement de-fenders	2.00
Joe Gomez	Liverpool	DF	Low attacking involvement de-fenders	5.67
John Stones	Manchester City	DF	Low attacking involvement de-fenders	9.96
Joël Veltman	Brighton	DF	Low attacking involvement de-fenders	4.56
Julian Araujo	Bournemouth	DF	Low attacking involvement de-fenders	3.64
Kevin Danso	Tottenham	DF	Low attacking involvement de-fenders	3.62
Konstantinos Mavropanos	West Ham	DF	Low attacking involvement de-fenders	3.24
Kostas Tsimikas	Liverpool	DF	Low attacking involvement de-fenders	6.15
Kristoffer Ajer	Brentford	DF	Low attacking involvement de-fenders	2.55
Kyle Walker	Manchester City	DF	Low attacking involvement de-fenders	4.22
Lamare Bogarde	Aston Villa	DF	Low attacking involvement de-fenders	5.84
Leny Yoro	Manchester Utd	DF	Low attacking involvement de-fenders	2.85
Levi Colwill	Chelsea	DF	Low attacking involvement de-fenders	2.38
Lewis Dunk	Brighton	DF	Low attacking involvement de-fenders	2.63
Luke Thomas	Leicester City	DF	Low attacking involvement de-fenders	4.26
Luke Woolfenden	Ipswich Town	DF	Low attacking involvement de-fenders	4.15
Manuel Akanji	Manchester City	DF	Low attacking involvement de-fenders	3.82
Marc Guéhi	Crystal Palace	DF	Low attacking involvement de-fenders	2.44
Marcos Senesi	Bournemouth	DF	Low attacking involvement de-fenders	4.07
Matt Doherty	Wolves	DF	Low attacking involvement de-fenders	1.89
Matthijs de Ligt	Manchester Utd	DF	Low attacking involvement de-fenders	4.15

Player	Club	Listed	Archetype	Distance
Matty Cash	Aston Villa	DF	Low attacking involvement de-fenders	4.01
Max Kilman	West Ham	DF	Low attacking involvement de-fenders	1.82
Maxence Lacroix	Crystal Palace	DF	Low attacking involvement de-fenders	2.59
Michael Keane	Everton	DF	Low attacking involvement de-fenders	5.83
Micky van de Ven	Tottenham	DF	Low attacking involvement de-fenders	3.68
Morato	Nottingham	DF	Low attacking involvement de-fenders	3.67
Murillo	Nottingham	DF	Low attacking involvement de-fenders	2.20
Myles Lewis-Skelly	Arsenal	DF	Low attacking involvement de-fenders	5.97
Nathan Aké	Manchester City	DF	Low attacking involvement de-fenders	3.94
Nathan Collins	Brentford	DF	Low attacking involvement de-fenders	2.35
Nathan Wood-Gordon	Southampton	DF	Low attacking involvement de-fenders	2.74
Nathaniel Clyne	Crystal Palace	DF	Low attacking involvement de-fenders	3.16
Nikola Milenković	Nottingham	DF	Low attacking involvement de-fenders	3.91
Noussair Mazraoui	Manchester Utd	DF	Low attacking involvement de-fenders	3.75
Ola Aina	Nottingham	DF	Low attacking involvement de-fenders	2.21
Pau Torres	Aston Villa	DF	Low attacking involvement de-fenders	2.70
Radu Drăgușin	Tottenham	DF	Low attacking involvement de-fenders	2.49
Rico Lewis	Manchester City	DF	Low attacking involvement de-fenders	4.39
Rúben Dias	Manchester City	DF	Low attacking involvement de-fenders	3.75
Santiago Bueno	Wolves	DF	Low attacking involvement de-fenders	2.97
Sepp van den Berg	Brentford	DF	Low attacking involvement de-fenders	2.18
Taylor Harwood-Bellis	Southampton	DF	Low attacking involvement de-fenders	2.28
Timothy Castagne	Fulham	DF	Low attacking involvement de-fenders	2.93
Tino Livramento	Newcastle	DF	Low attacking involvement de-fenders	3.48
Tosin Adarabioyo	Chelsea	DF	Low attacking involvement de-fenders	3.27
Toti Gomes	Wolves	DF	Low attacking involvement de-fenders	2.24

Player	Club	Listed	Archetype	Distance
Tyrone Mings	Aston Villa	DF	Low attacking involvement de-fenders	3.08
Victor Bernth Kristiansen	Leicester City	DF	Low attacking involvement de-fenders	4.22
Victor Lindelöf	Manchester Utd	DF	Low attacking involvement de-fenders	3.45
Virgil van Dijk	Liverpool	DF	Low attacking involvement de-fenders	4.08
Vitaliy Mykolenko	Everton	DF	Low attacking involvement de-fenders	3.14
Vladimír Coufal	West Ham	DF	Low attacking involvement de-fenders	4.23
Wesley Fofana	Chelsea	DF	Low attacking involvement de-fenders	4.97
William Saliba	Arsenal	DF	Low attacking involvement de-fenders	3.40
Wout Faes	Leicester City	DF	Low attacking involvement de-fenders	2.33
Álex Moreno	Nottingham	DF	Low attacking involvement de-fenders	3.76
Adam Armstrong	Southampton	FW	Crossing and creating wide forwards	3.69
Anthony Gordon	Newcastle	FW	Crossing and creating wide forwards	2.96
Bukayo Saka	Arsenal	FW	Crossing and creating wide forwards	6.23
Ethan Nwaneri	Arsenal	FW	Crossing and creating wide forwards	2.98
Fabio Carvalho	Brentford	FW	Crossing and creating wide forwards	5.32
Gabriel Martinelli	Arsenal	FW	Crossing and creating wide forwards	2.17
Harvey Barnes	Newcastle	FW	Crossing and creating wide forwards	2.69
Jacob Murphy	Newcastle	FW	Crossing and creating wide forwards	3.43
Joshua Zirkzee	Manchester Utd	FW	Crossing and creating wide forwards	3.12
João Pedro	Brighton	FW	Crossing and creating wide forwards	3.29
Kamaldeen Sulemana	Southampton	FW	Crossing and creating wide forwards	4.64
Leandro Trossard	Arsenal	FW	Crossing and creating wide forwards	1.28
Luis Díaz	Liverpool	FW	Crossing and creating wide forwards	4.40
Mathys Tel	Tottenham	FW	Crossing and creating wide forwards	2.45
Michail Antonio	West Ham	FW	Crossing and creating wide forwards	3.54
Raheem Sterling	Arsenal	FW	Crossing and creating wide forwards	5.73

Player	Club	Listed	Archetype	Distance
Son Heung-min	Tottenham	FW	Crossing and creating wide forwards	3.42
Timo Werner	Tottenham	FW	Crossing and creating wide forwards	5.35
Wilson Odobert	Tottenham	FW	Crossing and creating wide forwards	4.20
Alexander Isak	Newcastle	FW	Shot taking central forwards	3.60
Beto	Everton	FW	Shot taking central forwards	2.05
Brennan Johnson	Tottenham	FW	Shot taking central forwards	2.55
Cameron Archer	Southampton	FW	Shot taking central forwards	4.03
Chris Wood	Nottingham	FW	Shot taking central forwards	3.64
Danny Welbeck	Brighton	FW	Shot taking central forwards	2.85
Darwin Núñez	Liverpool	FW	Shot taking central forwards	5.20
Diogo Jota	Liverpool	FW	Shot taking central forwards	5.19
Dominic Calvert-Lewin	Everton	FW	Shot taking central forwards	4.01
Dominic Solanke	Tottenham	FW	Shot taking central forwards	2.44
Eddie Nketiah	Crystal Palace	FW	Shot taking central forwards	4.57
Erling Haaland	Manchester City	FW	Shot taking central forwards	4.19
Evanilson	Bournemouth	FW	Shot taking central forwards	2.19
Gabriel Jesus	Arsenal	FW	Shot taking central forwards	5.68
George Hirst	Ipswich Town	FW	Shot taking central forwards	2.57
Jamie Vardy	Leicester City	FW	Shot taking central forwards	3.08
Jean-Philippe Mateta	Crystal Palace	FW	Shot taking central forwards	2.09
Jáder Durán	Aston Villa	FW	Shot taking central forwards	7.28
Jørgen Strand Larsen	Wolves	FW	Shot taking central forwards	3.19
Kai Havertz	Arsenal	FW	Shot taking central forwards	1.80
Liam Delap	Ipswich Town	FW	Shot taking central forwards	2.81
Niclas Füllkrug	West Ham	FW	Shot taking central forwards	4.18
Nicolas Jackson	Chelsea	FW	Shot taking central forwards	1.92
Ollie Watkins	Aston Villa	FW	Shot taking central forwards	2.30
Paul Onuachu	Southampton	FW	Shot taking central forwards	2.75
Rasmus Højlund	Manchester Utd	FW	Shot taking central forwards	4.07
Raúl Jiménez	Fulham	FW	Shot taking central forwards	2.60
Richarlison	Tottenham	FW	Shot taking central forwards	3.79
Rodrigo Muniz	Fulham	FW	Shot taking central forwards	4.35
Yoane Wissa	Brentford	FW	Shot taking central forwards	2.27
Abdoulaye Doucouré	Everton	MF	Ball winning, low attacking involvement midfielders	3.11
Adam Wharton	Crystal Palace	MF	Ball winning, low attacking involvement midfielders	3.29
Alex Scott	Bournemouth	MF	Ball winning, low attacking involvement midfielders	4.20
Amadou Onana	Aston Villa	MF	Ball winning, low attacking involvement midfielders	3.37
André	Wolves	MF	Ball winning, low attacking involvement midfielders	2.86
Ben Johnson	Ipswich Town	MF	Ball winning, low attacking involvement midfielders	2.89
Bilal El Khannouss	Leicester City	MF	Ball winning, low attacking involvement midfielders	3.33

Player	Club	Listed	Archetype	Distance
Bobby De Cordova-Reid	Leicester City	MF	Ball winning, low attacking involvement midfielders	4.65
Boubacar Kamara	Aston Villa	MF	Ball winning, low attacking involvement midfielders	2.30
Boubakary Soumaré	Leicester City	MF	Ball winning, low attacking involvement midfielders	3.36
Bruno Guimarães	Newcastle	MF	Ball winning, low attacking involvement midfielders	3.65
Carlos Baleba	Brighton	MF	Ball winning, low attacking involvement midfielders	2.81
Carlos Soler	West Ham	MF	Ball winning, low attacking involvement midfielders	2.53
Casemiro	Manchester Utd	MF	Ball winning, low attacking involvement midfielders	4.01
Cheick Doucouaré	Crystal Palace	MF	Ball winning, low attacking involvement midfielders	5.22
Christian Nørgaard	Brentford	MF	Ball winning, low attacking involvement midfielders	2.64
Conor Chaplin	Ipswich Town	MF	Ball winning, low attacking involvement midfielders	2.92
Daichi Kamada	Crystal Palace	MF	Ball winning, low attacking involvement midfielders	1.75
Daniel Muñoz	Crystal Palace	MF	Ball winning, low attacking involvement midfielders	3.51
Diogo Dalot	Manchester Utd	MF	Ball winning, low attacking involvement midfielders	2.19
Edson Álvarez	West Ham	MF	Ball winning, low attacking involvement midfielders	3.08
Elliot Anderson	Nottingham	MF	Ball winning, low attacking involvement midfielders	2.18
Flynn Downes	Southampton	MF	Ball winning, low attacking involvement midfielders	2.78
Guido Rodríguez	West Ham	MF	Ball winning, low attacking involvement midfielders	2.80
Harry Winks	Leicester City	MF	Ball winning, low attacking involvement midfielders	3.17
Hwang Hee-chan	Wolves	MF	Ball winning, low attacking involvement midfielders	6.60
Ibrahim Sangaré	Nottingham	MF	Ball winning, low attacking involvement midfielders	6.14
Idrissa Gueye	Everton	MF	Ball winning, low attacking involvement midfielders	2.34
Jack Clarke	Ipswich Town	MF	Ball winning, low attacking involvement midfielders	4.03
Jack Harrison	Everton	MF	Ball winning, low attacking involvement midfielders	4.30
Jack Hinshelwood	Brighton	MF	Ball winning, low attacking involvement midfielders	4.61
Jack Taylor	Ipswich Town	MF	Ball winning, low attacking involvement midfielders	2.43
Jacob Ramsey	Aston Villa	MF	Ball winning, low attacking involvement midfielders	2.84

Player	Club	Listed	Archetype	Distance
Jaden Philogene-Bidace	Ipswich Town / Aston Villa	MF	Ball winning, low attacking involvement midfielders	4.28
James Garner	Everton	MF	Ball winning, low attacking involvement midfielders	2.04
James Ward-Prowse	West Ham / Nottingham	MF	Ball winning, low attacking involvement midfielders	3.03
Jean-Ricner Bellegarde	Wolves	MF	Ball winning, low attacking involvement midfielders	4.06
Jefferson Lerma	Crystal Palace	MF	Ball winning, low attacking involvement midfielders	2.61
Jens Cajuste	Ipswich Town	MF	Ball winning, low attacking involvement midfielders	2.83
Jesper Lindstrøm	Everton	MF	Ball winning, low attacking involvement midfielders	3.68
Joe Aribo	Southampton	MF	Ball winning, low attacking involvement midfielders	3.13
Joe Willock	Newcastle	MF	Ball winning, low attacking involvement midfielders	3.56
Joelinton	Newcastle	MF	Ball winning, low attacking involvement midfielders	3.01
John McGinn	Aston Villa	MF	Ball winning, low attacking involvement midfielders	3.29
Jordan Ayew	Leicester City / Crystal Palace	MF	Ball winning, low attacking involvement midfielders	4.60
Jorginho	Arsenal	MF	Ball winning, low attacking involvement midfielders	4.85
João Gomes	Wolves	MF	Ball winning, low attacking involvement midfielders	2.56
Kalvin Phillips	Ipswich Town	MF	Ball winning, low attacking involvement midfielders	2.81
Kasey McAteer	Leicester City	MF	Ball winning, low attacking involvement midfielders	3.50
Kobbie Mainoo	Manchester Utd	MF	Ball winning, low attacking involvement midfielders	1.93
Kyle Walker-Peters	Southampton	MF	Ball winning, low attacking involvement midfielders	2.29
Lesley Ugochukwu	Southampton	MF	Ball winning, low attacking involvement midfielders	3.93
Lewis Cook	Bournemouth	MF	Ball winning, low attacking involvement midfielders	2.67
Lucas Bergvall	Tottenham	MF	Ball winning, low attacking involvement midfielders	2.27
Lucas Paquetá	West Ham	MF	Ball winning, low attacking involvement midfielders	2.85
Manuel Ugarte	Manchester Utd	MF	Ball winning, low attacking involvement midfielders	3.67
Mario Lemina	Wolves	MF	Ball winning, low attacking involvement midfielders	3.07
Mateo Kovačić	Manchester City	MF	Ball winning, low attacking involvement midfielders	3.82
Mateus Fernandes	Southampton	MF	Ball winning, low attacking involvement midfielders	1.86

Player	Club	Listed	Archetype	Distance
Mathias Jensen	Brentford	MF	Ball winning, low attacking involvement midfielders	3.67
Moisés Caicedo	Chelsea	MF	Ball winning, low attacking involvement midfielders	3.08
Nicolás Domínguez	Nottingham	MF	Ball winning, low attacking involvement midfielders	2.76
Nicolás González	Manchester City	MF	Ball winning, low attacking involvement midfielders	5.07
Nélson Semedo	Wolves	MF	Ball winning, low attacking involvement midfielders	1.85
Oliver Scarles	West Ham	MF	Ball winning, low attacking involvement midfielders	4.31
Oliver Skipp	Leicester City	MF	Ball winning, low attacking involvement midfielders	4.02
Omari Hutchinson	Ipswich Town	MF	Ball winning, low attacking involvement midfielders	3.30
Orel Mangala	Everton	MF	Ball winning, low attacking involvement midfielders	2.87
Pape Matar Sarr	Tottenham	MF	Ball winning, low attacking involvement midfielders	2.51
Patrick Dorgu	Manchester Utd	MF	Ball winning, low attacking involvement midfielders	4.54
Patson Daka	Leicester City	MF	Ball winning, low attacking involvement midfielders	4.79
Rayan Aït-Nouri	Wolves	MF	Ball winning, low attacking involvement midfielders	2.51
Rodrigo Bentancur	Tottenham	MF	Ball winning, low attacking involvement midfielders	4.07
Rodrigo Gomes	Wolves	MF	Ball winning, low attacking involvement midfielders	5.18
Roméo Lavia	Chelsea	MF	Ball winning, low attacking involvement midfielders	4.01
Ryan Christie	Bournemouth	MF	Ball winning, low attacking involvement midfielders	2.35
Ryan Gravenberch	Liverpool	MF	Ball winning, low attacking involvement midfielders	3.56
Ryan Yates	Nottingham	MF	Ball winning, low attacking involvement midfielders	3.48
Sam Morsy	Ipswich Town	MF	Ball winning, low attacking involvement midfielders	1.72
Sander Berge	Fulham	MF	Ball winning, low attacking involvement midfielders	2.59
Sandro Tonali	Newcastle	MF	Ball winning, low attacking involvement midfielders	2.14
Saša Lukić	Fulham	MF	Ball winning, low attacking involvement midfielders	2.97
Sean Longstaff	Newcastle	MF	Ball winning, low attacking involvement midfielders	2.57
Thomas Partey	Arsenal	MF	Ball winning, low attacking involvement midfielders	2.90
Tim Iroegbunam	Everton	MF	Ball winning, low attacking involvement midfielders	4.93

Player	Club	Listed	Archetype	Distance
Tommy Doyle	Wolves	MF	Ball winning, low attacking involvement midfielders	2.40
Tomáš Souček	West Ham	MF	Ball winning, low attacking involvement midfielders	3.81
Tyler Adams	Bournemouth	MF	Ball winning, low attacking involvement midfielders	3.23
Tyler Dibling	Southampton	MF	Ball winning, low attacking involvement midfielders	2.69
Tyrick Mitchell	Crystal Palace	MF	Ball winning, low attacking involvement midfielders	3.31
Vitaly Janelt	Brentford	MF	Ball winning, low attacking involvement midfielders	2.17
Welington	Southampton	MF	Ball winning, low attacking involvement midfielders	3.86
Wes Burns	Ipswich Town	MF	Ball winning, low attacking involvement midfielders	3.86
Wilfred Ndidi	Leicester City	MF	Ball winning, low attacking involvement midfielders	2.10
Will Hughes	Crystal Palace	MF	Ball winning, low attacking involvement midfielders	3.68
William Smallbone	Southampton	MF	Ball winning, low attacking involvement midfielders	3.41
Yasin Ayari	Brighton	MF	Ball winning, low attacking involvement midfielders	1.95
Yehor Yarmoliuk	Brentford	MF	Ball winning, low attacking involvement midfielders	2.16
Yukinari Sugawara	Southampton	MF	Ball winning, low attacking involvement midfielders	3.10
Yves Bissouma	Tottenham	MF	Ball winning, low attacking involvement midfielders	3.20
Abdul Fatawu Issahaku	Leicester City	MF	High attacking involvement midfielders	5.45
Adama Traoré	Fulham	MF	High attacking involvement midfielders	3.33
Alejandro Garnacho	Manchester Utd	MF	High attacking involvement midfielders	3.19
Alex Iwobi	Fulham	MF	High attacking involvement midfielders	2.34
Alexis Mac Allister	Liverpool	MF	High attacking involvement midfielders	4.79
Amad Diallo	Manchester Utd	MF	High attacking involvement midfielders	2.55
Andreas Pereira	Fulham	MF	High attacking involvement midfielders	4.18
Anthony Elanga	Nottingham	MF	High attacking involvement midfielders	4.70
Antoine Semenyo	Bournemouth	MF	High attacking involvement midfielders	3.40
Bernardo Silva	Manchester City	MF	High attacking involvement midfielders	2.96
Brajan Gruda	Brighton	MF	High attacking involvement midfielders	4.51

Player	Club	Listed	Archetype	Distance
Bruno Fernandes	Manchester Utd	MF	High attacking involvement mid-fielders	3.45
Bryan Mbeumo	Brentford	MF	High attacking involvement mid-fielders	2.94
Callum Hudson-Odoi	Nottingham	MF	High attacking involvement mid-fielders	2.94
Carlos Alcaraz	Everton / Southampton	MF	High attacking involvement mid-fielders	4.16
Christian Eriksen	Manchester Utd	MF	High attacking involvement mid-fielders	5.66
Christopher Nkunku	Chelsea	MF	High attacking involvement mid-fielders	3.16
Cody Gakpo	Liverpool	MF	High attacking involvement mid-fielders	3.23
Cole Palmer	Chelsea	MF	High attacking involvement mid-fielders	3.63
Crysencio Summerville	West Ham	MF	High attacking involvement mid-fielders	3.52
Curtis Jones	Liverpool	MF	High attacking involvement mid-fielders	2.94
Dango Ouattara	Bournemouth	MF	High attacking involvement mid-fielders	3.50
David Brooks	Bournemouth	MF	High attacking involvement mid-fielders	3.02
Declan Rice	Arsenal	MF	High attacking involvement mid-fielders	3.32
Dejan Kulusevski	Tottenham	MF	High attacking involvement mid-fielders	2.19
Diego Gómez	Brighton	MF	High attacking involvement mid-fielders	5.42
Dominik Szoboszlai	Liverpool	MF	High attacking involvement mid-fielders	3.31
Dwight McNeil	Everton	MF	High attacking involvement mid-fielders	4.49
Eberechi Eze	Crystal Palace	MF	High attacking involvement mid-fielders	2.84
Emile Smith Rowe	Fulham	MF	High attacking involvement mid-fielders	3.02
Enzo Fernández	Chelsea	MF	High attacking involvement mid-fielders	2.23
Facundo Buonanotte	Leicester City	MF	High attacking involvement mid-fielders	3.85
Georginio Rutter	Brighton	MF	High attacking involvement mid-fielders	3.59
Gonçalo Guedes	Wolves	MF	High attacking involvement mid-fielders	2.90
Harry Wilson	Fulham	MF	High attacking involvement mid-fielders	3.33
Iliman Ndiaye	Everton	MF	High attacking involvement mid-fielders	4.21
Ismaila Sarr	Crystal Palace	MF	High attacking involvement mid-fielders	3.11

Player	Club	Listed	Archetype	Distance
Jack Grealish	Manchester City	MF	High attacking involvement mid-fielders	4.82
Jadon Sancho	Chelsea	MF	High attacking involvement mid-fielders	3.50
James Maddison	Tottenham	MF	High attacking involvement mid-fielders	4.35
Jarrod Bowen	West Ham	MF	High attacking involvement mid-fielders	2.76
Jeremy Doku	Manchester City	MF	High attacking involvement mid-fielders	3.11
Jota Silva	Nottingham	MF	High attacking involvement mid-fielders	6.09
Julio Enciso	Ipswich Town / Brighton	MF	High attacking involvement mid-fielders	4.26
Justin Devenny	Crystal Palace	MF	High attacking involvement mid-fielders	4.23
Justin Kluivert	Bournemouth	MF	High attacking involvement mid-fielders	2.37
Kaoru Mitoma	Brighton	MF	High attacking involvement mid-fielders	2.46
Kevin De Bruyne	Manchester City	MF	High attacking involvement mid-fielders	5.25
Kevin Schade	Brentford	MF	High attacking involvement mid-fielders	3.76
Leon Bailey	Aston Villa	MF	High attacking involvement mid-fielders	3.04
Marco Asensio	Aston Villa	MF	High attacking involvement mid-fielders	2.77
Marcus Rashford	Manchester Utd / Aston Villa	MF	High attacking involvement mid-fielders	2.76
Marcus Tavernier	Bournemouth	MF	High attacking involvement mid-fielders	2.47
Marshall Munetsi	Wolves	MF	High attacking involvement mid-fielders	3.48
Martin Ødegaard	Arsenal	MF	High attacking involvement mid-fielders	3.56
Mason Mount	Manchester Utd	MF	High attacking involvement mid-fielders	4.72
Matheus Cunha	Wolves	MF	High attacking involvement mid-fielders	3.96
Matt O'Riley	Brighton	MF	High attacking involvement mid-fielders	2.71
Mikel Merino	Arsenal	MF	High attacking involvement mid-fielders	4.28
Mikkel Damsgaard	Brentford	MF	High attacking involvement mid-fielders	4.10
Mohamed Salah	Liverpool	MF	High attacking involvement mid-fielders	7.03
Mohammed Kudus	West Ham	MF	High attacking involvement mid-fielders	3.60
Morgan Gibbs-White	Nottingham	MF	High attacking involvement mid-fielders	2.01

Player	Club	Listed	Archetype	Distance
Morgan Rogers	Aston Villa	MF	High attacking involvement mid-fielders	2.65
Nathan Broadhead	Ipswich Town	MF	High attacking involvement mid-fielders	3.77
Noni Madueke	Chelsea	MF	High attacking involvement mid-fielders	3.54
Omar Marmoush	Manchester City	MF	High attacking involvement mid-fielders	5.31
Pablo Sarabia	Wolves	MF	High attacking involvement mid-fielders	3.09
Pedro Neto	Chelsea	MF	High attacking involvement mid-fielders	2.78
Phil Foden	Manchester City	MF	High attacking involvement mid-fielders	2.92
Reiss Nelson	Fulham / Arsenal	MF	High attacking involvement mid-fielders	3.23
Ross Barkley	Aston Villa	MF	High attacking involvement mid-fielders	5.01
Ryan Sessegnon	Fulham	MF	High attacking involvement mid-fielders	6.16
Sammie Szmodics	Ipswich Town	MF	High attacking involvement mid-fielders	5.68
Simon Adingra	Brighton	MF	High attacking involvement mid-fielders	3.23
Stephy Mavididi	Leicester City	MF	High attacking involvement mid-fielders	2.81
Sávio	Manchester City	MF	High attacking involvement mid-fielders	3.64
Tom Cairney	Fulham	MF	High attacking involvement mid-fielders	5.50
Yankuba Minteh	Brighton	MF	High attacking involvement mid-fielders	2.81
Youri Tielemans	Aston Villa	MF	High attacking involvement mid-fielders	3.65
İlkay Gündoğan	Manchester City	MF	High attacking involvement mid-fielders	4.37

F Supplementary figures

Figure 25 gives the full hierarchically clustered Spearman correlation matrix over the outfield feature set, from which the divergence analysis in Section 4.5 is derived. Figure 26a reports the scree and cumulative variance curves: 9 components are required to reach the ninety percent threshold, so the feature set is not close to low rank, and the first three components together account for 0.634 of the variance. Figure 26b shows the loadings biplot. Figure 27a gives the correlation structure of the surviving goalkeeper features and Figure 27b the distribution of the two goalkeeper composites.

G Individual archetype radars

Figures 28 to 30 give the archetype radars for each position group at full size, on the fixed axes described in Section 3. Within each position group the two profiles are mirror images by construction, since the features are standardised within group before clustering.

Shot-stopping index. Volume adjusted only, not shot-quality adjusted

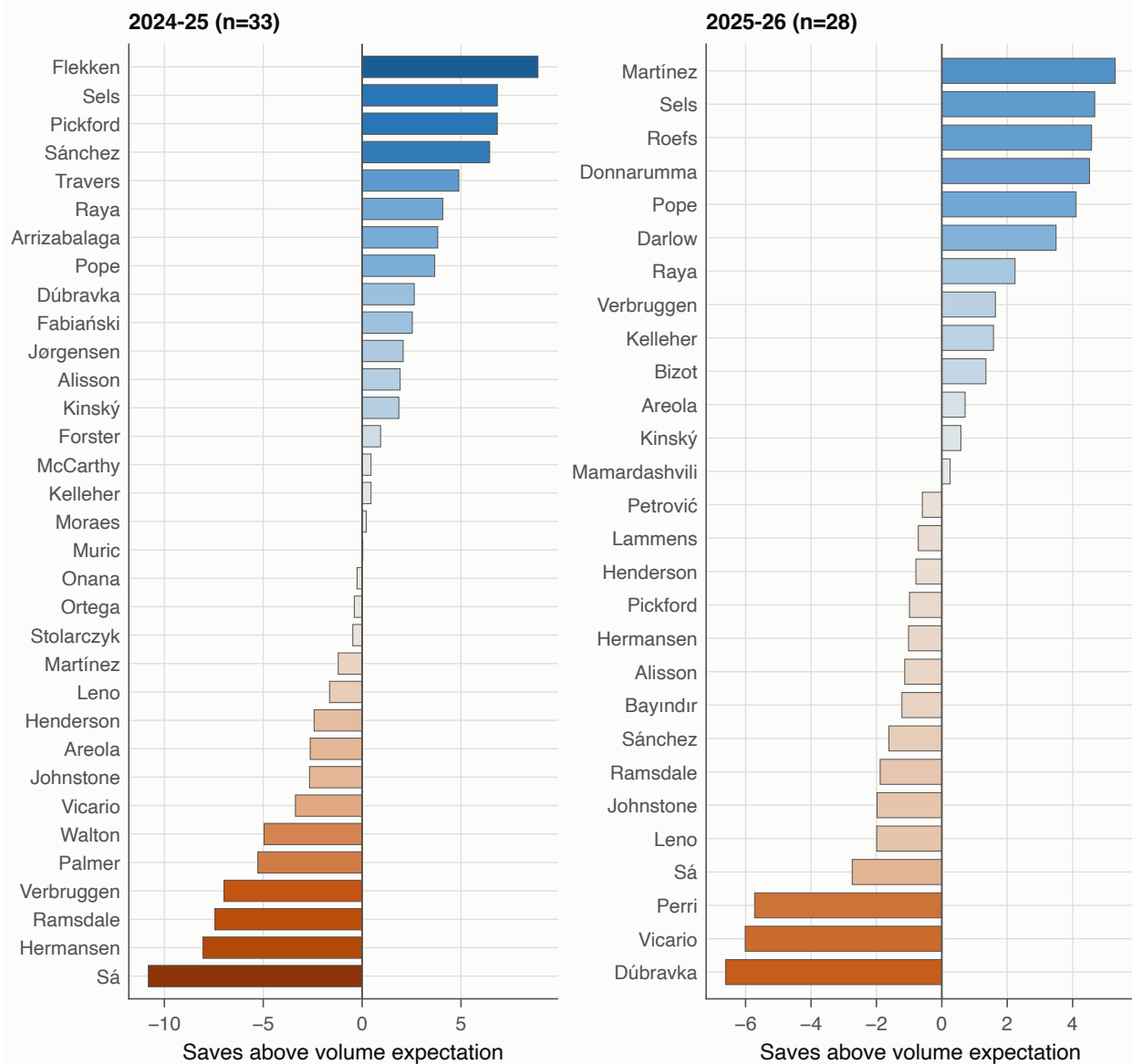
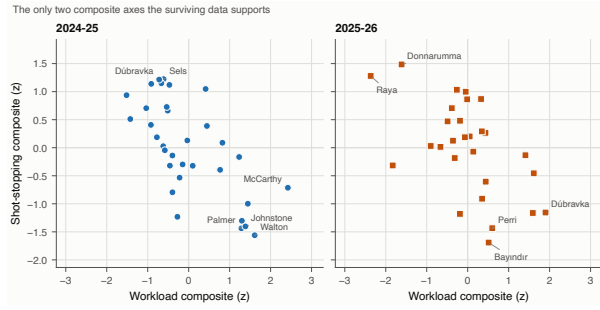
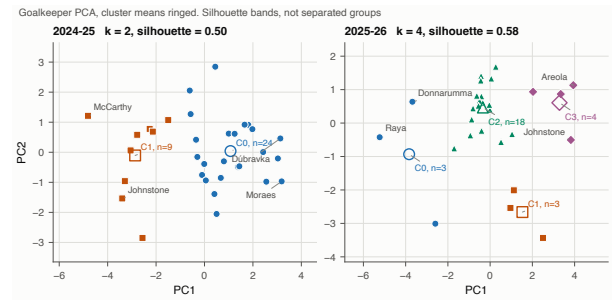


Figure 13: Goalkeepers ranked by saves above the volume of shots on target faced. The index adjusts for how much work a goalkeeper receives but not for its difficulty.



(a) The two surviving composite axes.



(b) The goalkeeper partition, shown for completeness.

Figure 14: Goalkeeper composites and the partition that the validation checks reject.

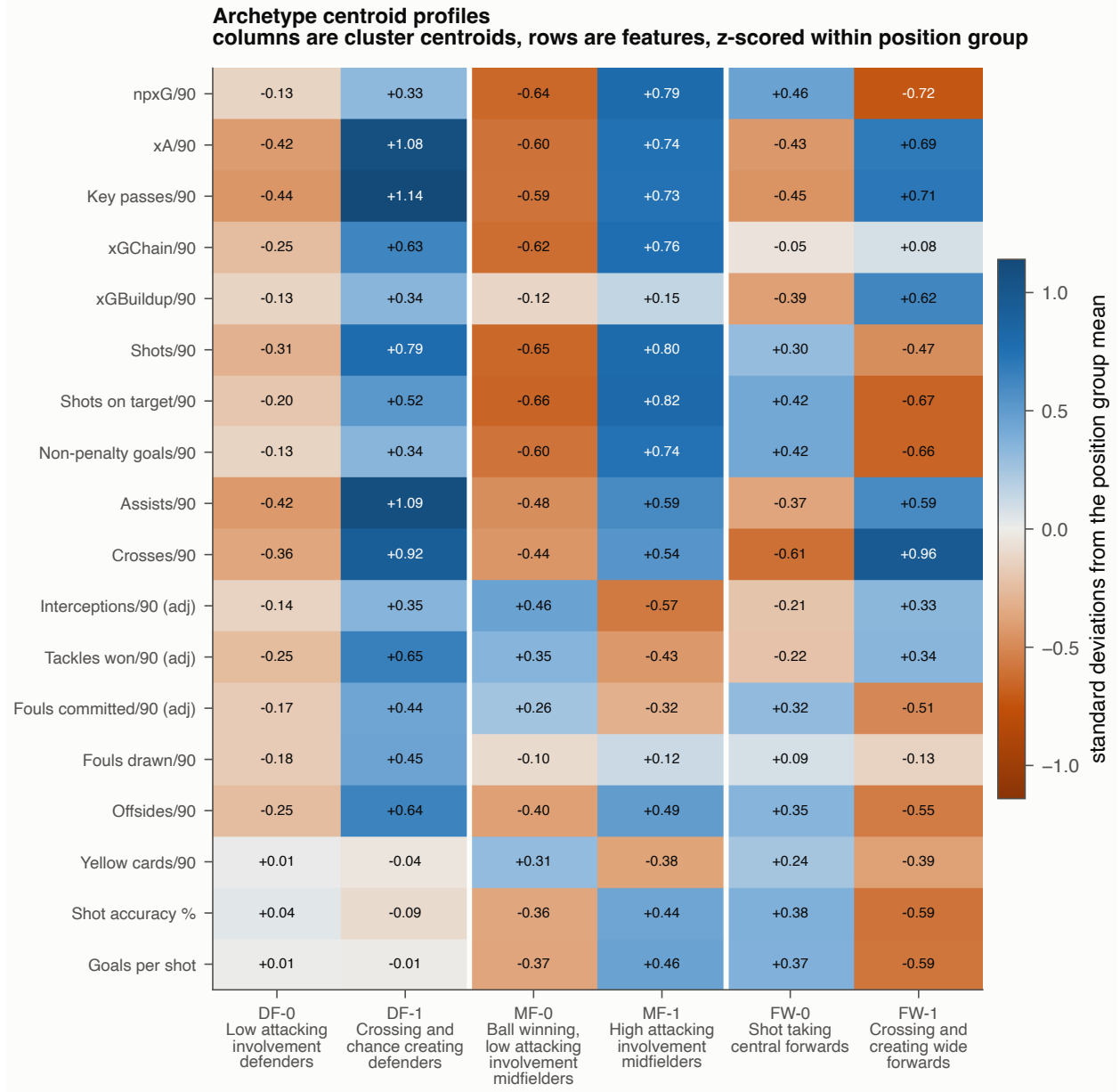


Figure 15: Centroid deviation of each archetype from its position group mean, across the full feature set. Within each position group the two profiles are necessarily mirror images, since the features are standardised within group before clustering.

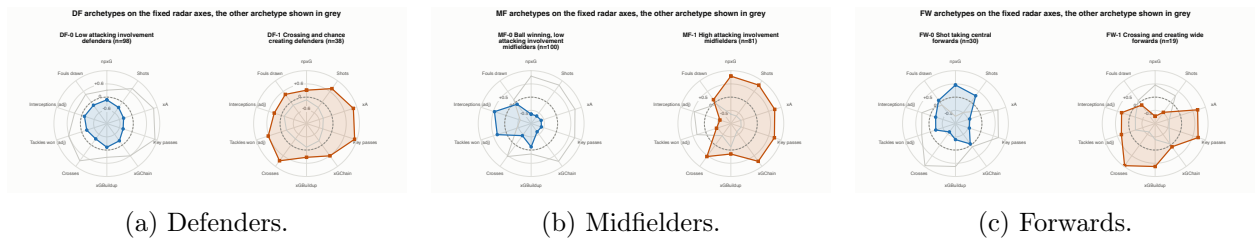
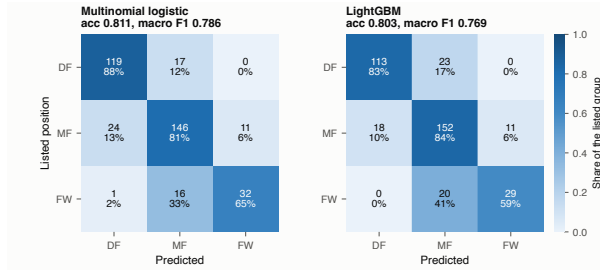
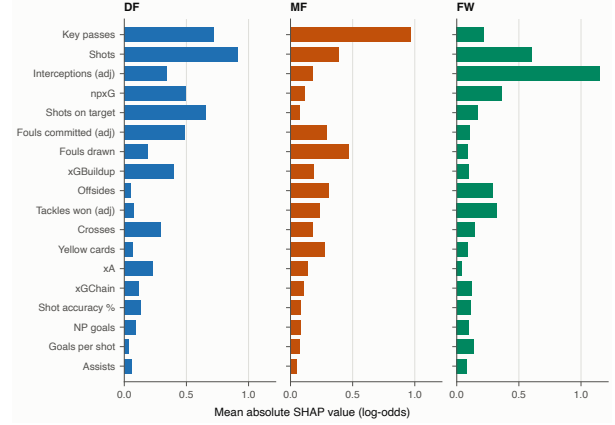


Figure 16: Archetype radars on fixed axes within each position group.

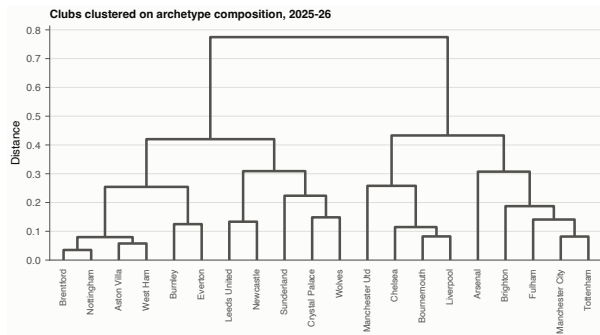


(a) Confusion between listed positions.

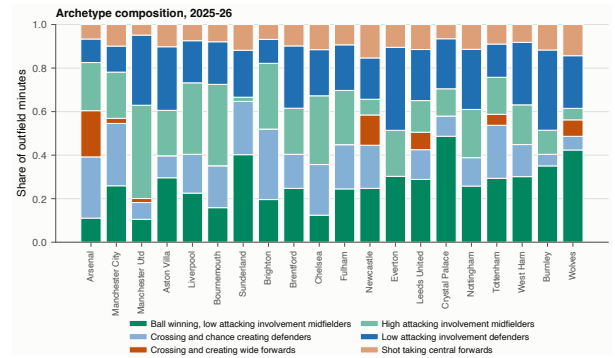


(b) Feature importance per position.

Figure 17: Supervised validation of the surviving feature set.



(a) Style families.

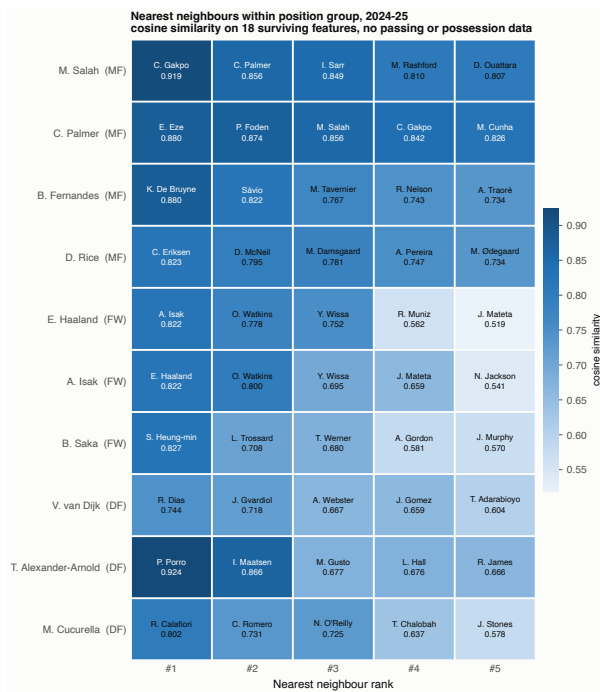


(b) Archetype composition by club.

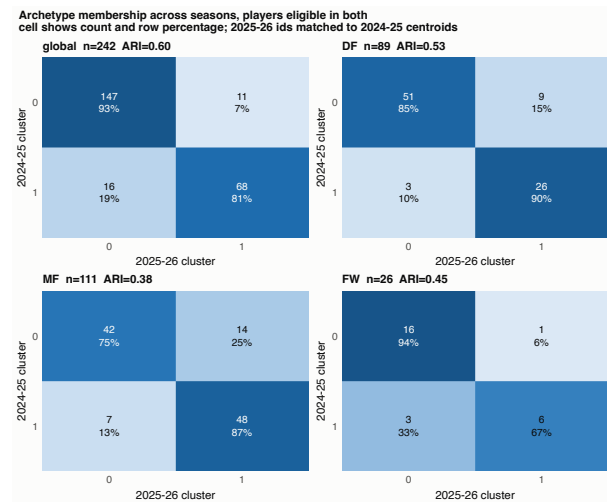
Figure 18: Club stylistic signatures, ordered by final league position.



Figure 19: Each club's outfield players highlighted in the shared embedding, with the rest of the league in grey. Axes are shared across all twenty panels.

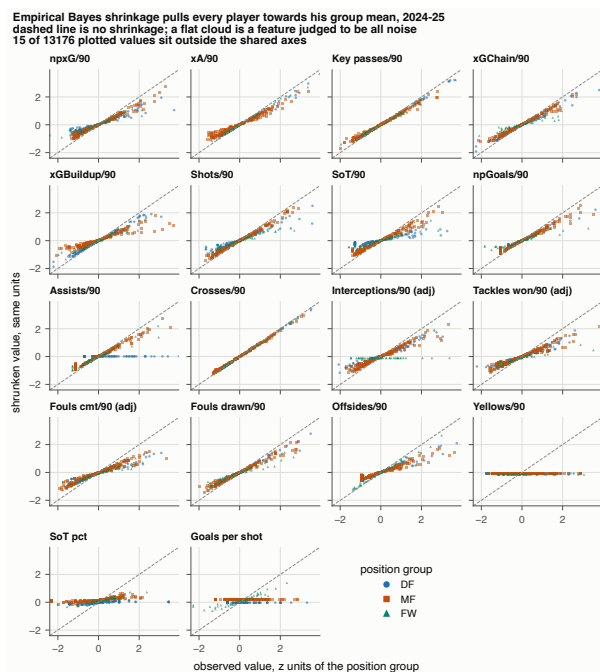


(a) Nearest neighbour similarity.

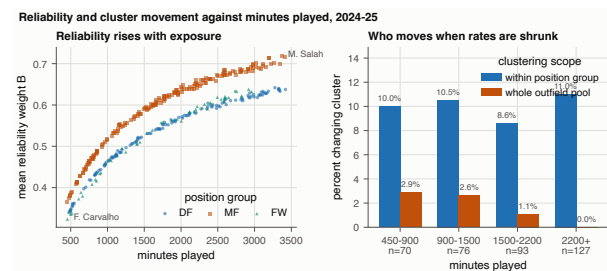


(b) Between-season transitions.

Figure 20: Similarity structure and cross-season replication.

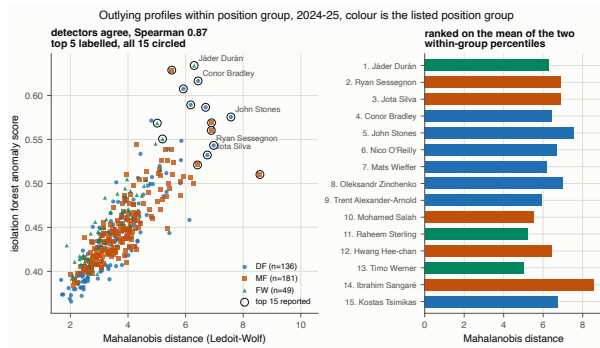


(a) Effect of shrinkage on assignments.

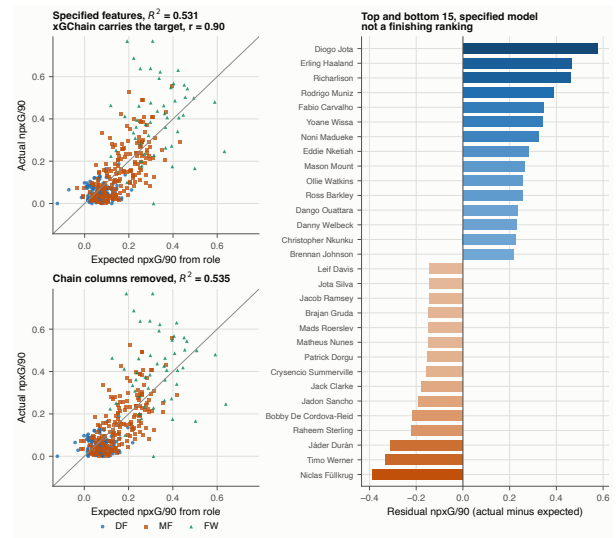


(b) Assignment changes by minutes played.

Figure 21: Empirical Bayes shrinkage as a test of whether the partitions describe roles or noise.



(a) Outliers within position group.



(b) Residuals from the role model.

Figure 22: Unusual players, and the residual leaderboard that should not be read as a measurement of finishing ability.

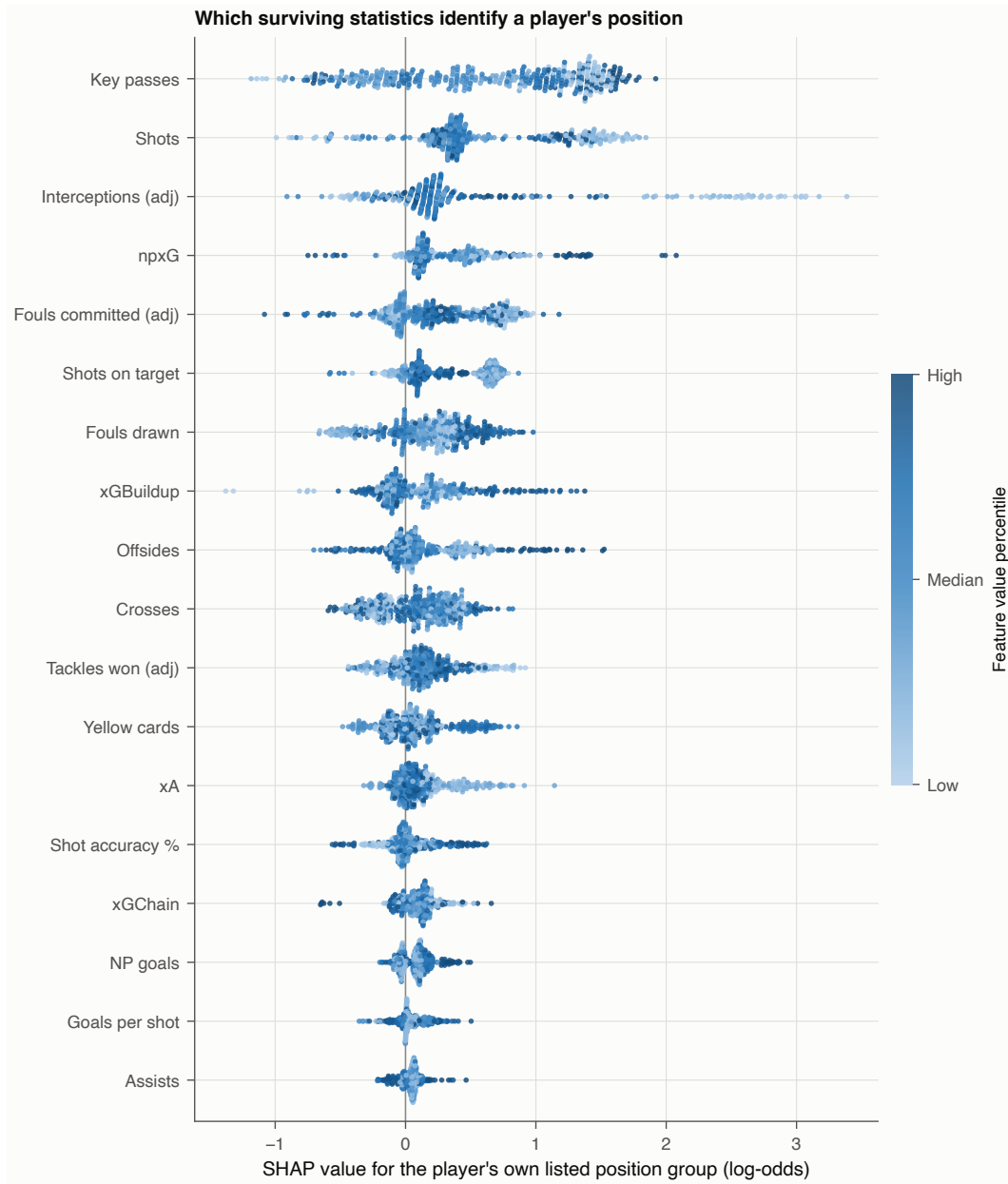


Figure 23: Global SHAP values for the gradient boosted position classifier. Shooting volume and interceptions carry most of the signal, so a player's position is now identified largely by the actions he does not perform.

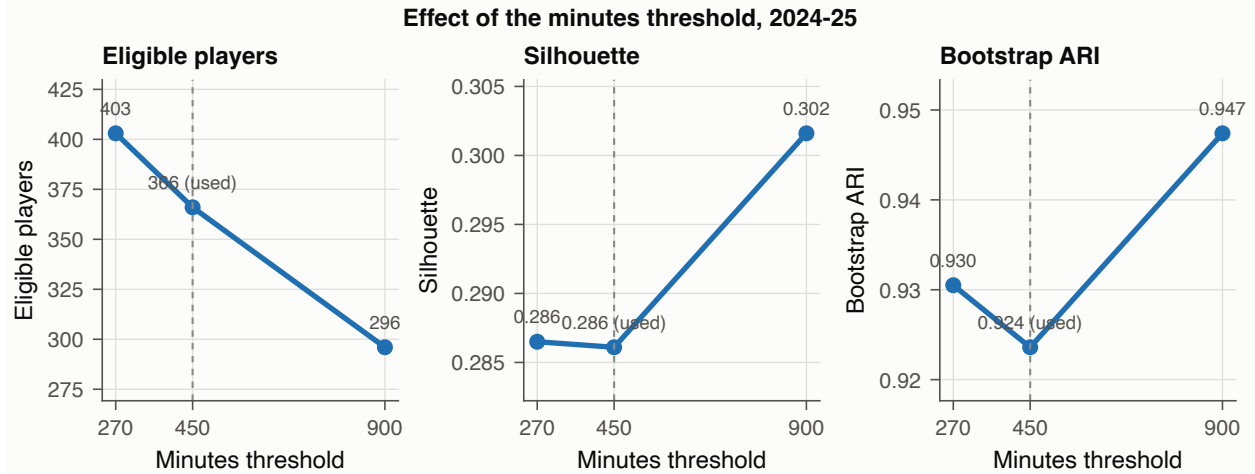


Figure 24: Eligible sample, silhouette and bootstrap stability across the candidate minutes thresholds. The dashed line marks the threshold used in the main analysis.

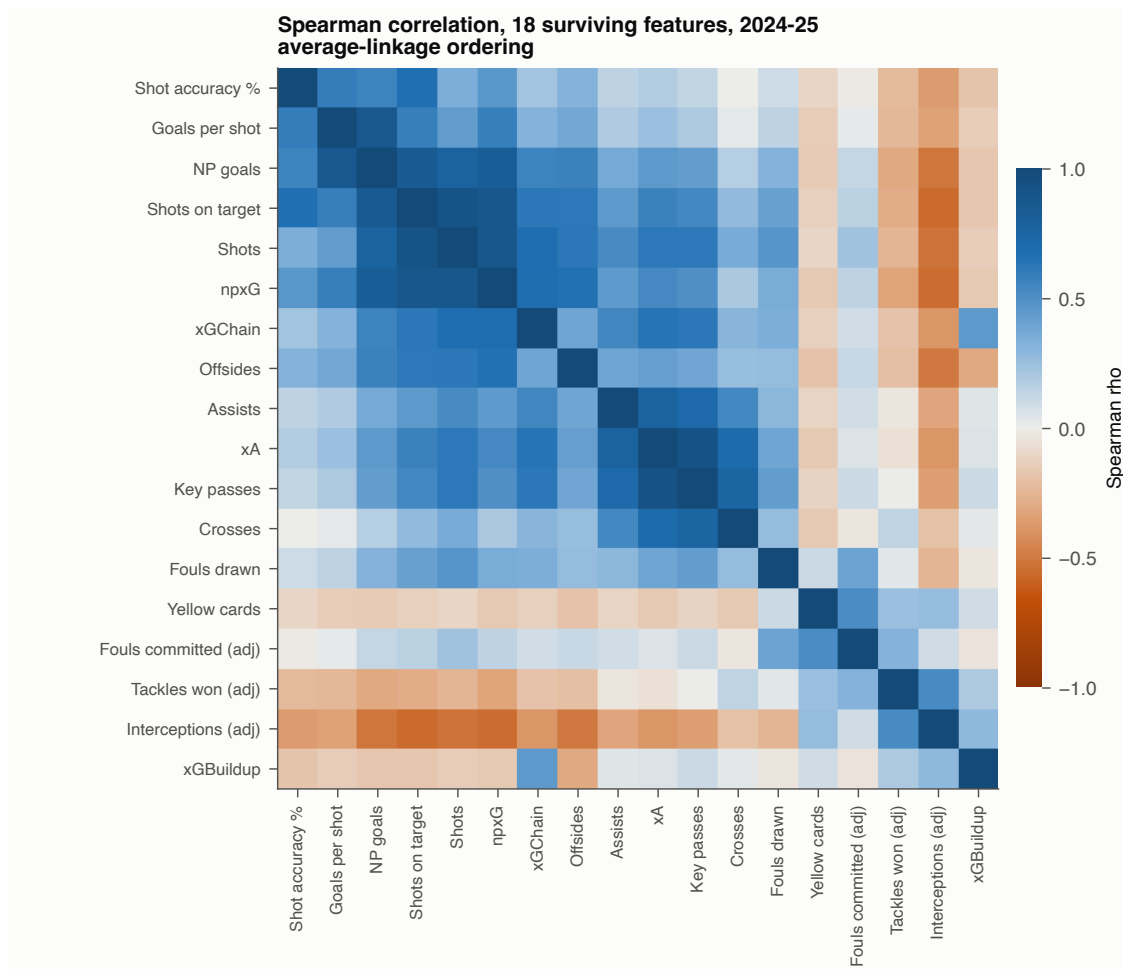
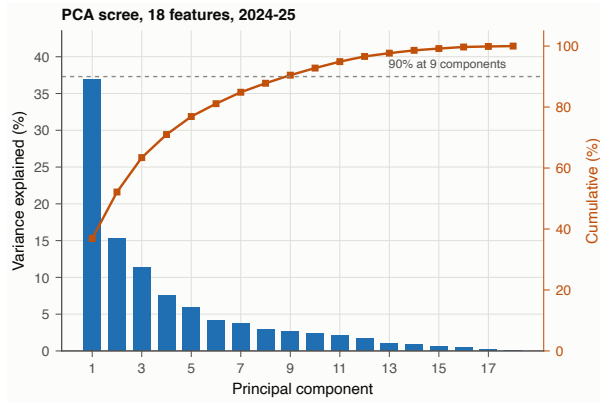
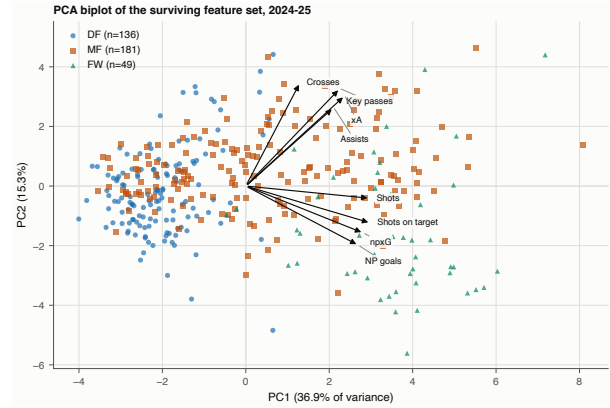


Figure 25: Hierarchically clustered Spearman correlation matrix over the outfield feature set in the primary season.

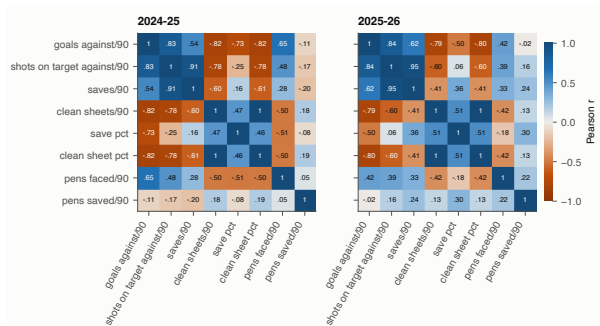


(a) Scree and cumulative variance.

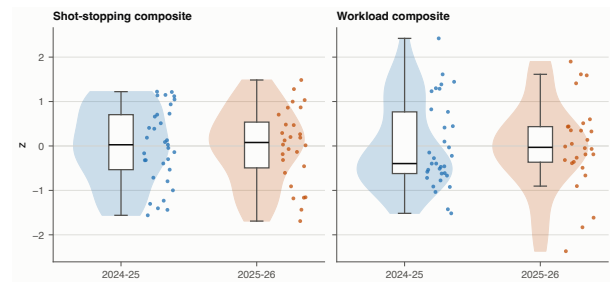


(b) Loadings biplot.

Figure 26: Principal component structure of the outfield feature set.



(a) Surviving goalkeeper features.



(b) Goalkeeper composite distributions.

Figure 27: Supplementary goalkeeper material.

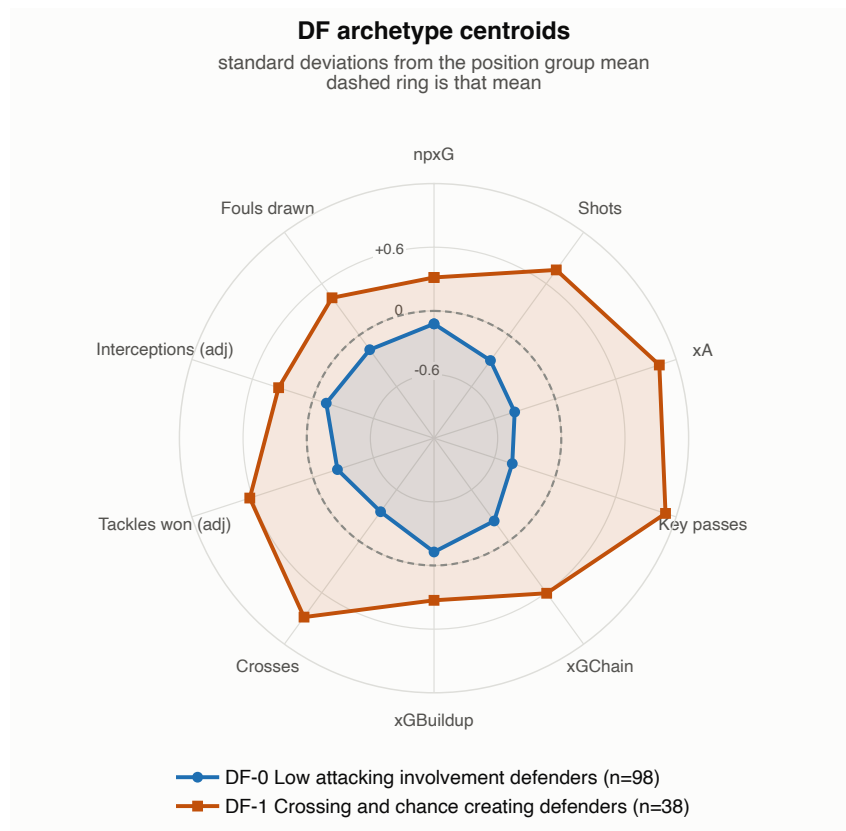


Figure 28: Defender archetypes.

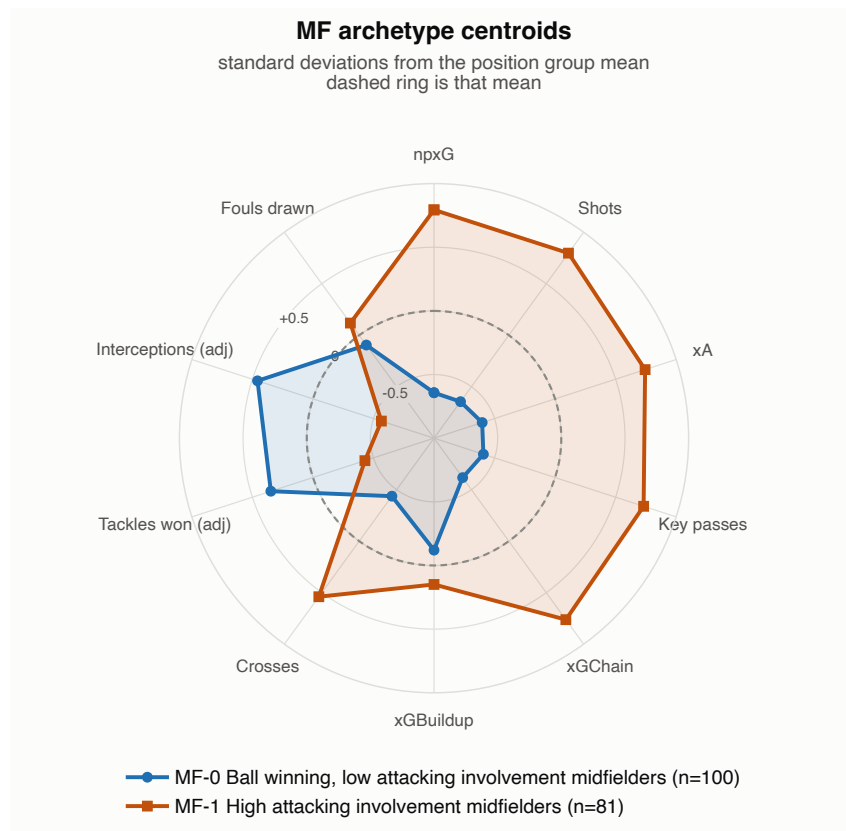


Figure 29: Midfielder archetypes.

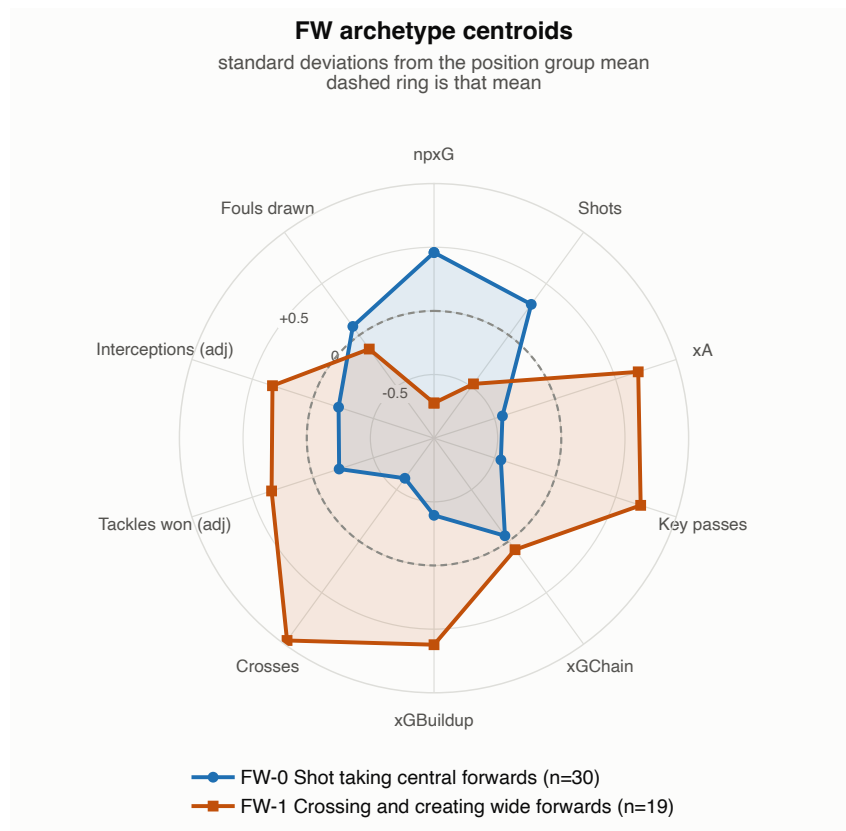


Figure 30: Forward archetypes.